Core Financial Modeling, Modules 9 – 10: M&A Deals and Merger Models

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Overview & Key Rules of Thumb

M&A deals and merger models are likely topics in finance interviews, but they're **less important** than accounting, valuation, and DCF analysis for several reasons:

1. Mergers and acquisitions are **not relevant for all groups**. For example, you don't work on M&A deals in the debt capital markets (DCM) or equity capital markets (ECM) groups in investment banking.

2. They're also **less important in roles** like equity research and hedge funds, where you follow public companies and make investment recommendations.

3. They're **more advanced** than the other topics, so they're less likely to be the subject of interview questions.

You'll get at least a few questions about M&A deals if you interview for investment banking roles, but advanced questions are unlikely unless you have significant work experience.

Basic M&A analysis is **simple** – for example, it's easier to explain than 3-statement projection models.

But the interview questions can be quite tricky since many candidates memorize questions and answers without understanding the *underlying concepts*.

We start this guide by describing **why** one company might want to buy another company, and then we move on to simple examples of merger models so you can learn the building blocks.

Then, we walk through a full merger model for real companies and other ways to evaluate deals besides the traditional EPS accretion/dilution analysis.

Key Rule #1: Why Buy Another Company?

A company might buy another company if it thinks that company is worth *more* than the price it's paying.

So, maybe the Target's asking price is $100, but the Acquirer thinks it's worth $120 or $150 based on its future cash flows.

Or maybe the Acquirer has a targeted rate of return of 10% on all its deals, and it thinks it can earn 15% per year by purchasing the Target.

Since the expected annualized return exceeds its targeted return, the deal works.

These rules based on cash flows and valuations make sense theoretically, but something else enters the equation in real life: **Earnings per Share (EPS)**.

A company's shareholders and Board of Directors care *a lot* about metrics such as Net Income and EPS – and how acquisitions might affect them.

These metrics differ from "cash flow," but many stocks still trade in the short term based on Net Income and EPS changes.

EPS is also important because it reflects *all* the effects of an acquisition: Foregone interest on cash, interest paid on new debt, and the new shares issued to fund the deal.

So, if a company announces an acquisition that will "boost its long-term cash flows," but it will also take a huge EPS hit next year, the market tends to react poorly.

Therefore, companies prefer acquisitions that are **accretive to EPS**; in other words, acquisitions that *increase* their EPS.

For example, if a company expects to earn $2.00 per share next year, an EPS-accretive deal might boost its EPS to $2.10 or $2.50.

A deal that's **dilutive to EPS** might reduce its EPS to $1.50 or $1.80.

These are the two most important **financial criteria** for deals: The price paid for the Target must be reasonable, and the deal should be neutral or accretive to EPS.

But in real life, no company *decides* to make a major acquisition based solely on valuation and EPS; they're more like screening tools or supporting data.

Common motivation to do deals include:

- **Consolidation / Economies of Scale:** If the biggest and second biggest companies in the market combine, they can get better deals with suppliers and save money. Or they can consolidate departments like accounting and HR, lay off employees, and save money.

- **Geographic Expansion:** The Buyer operates mostly in Europe, which is a declining/mature market. It wants to acquire the Seller to expand into Asia and grow more quickly.

- **Gain Market Share:** Neither the Buyer nor the Seller is growing because the market is competitive and highly fragmented, and different companies' products are similar. But if the Buyer acquires the Seller, it instantly captures market share.

- **Acquire Customers or Distribution Channels:** One company could sell more widgets if it had access to another company's customers, partners, and sales force.

- **Product Expansion or Diversification:** The Acquirer's core industry is declining, but it could grow more quickly by purchasing a company in a higher-growth industry.

- **Intellectual Property:** The Acquirer wants a cool, shiny thing that the Target has. It can't determine how much revenue, profit, or cash flow this shiny thing will produce…. but it's shiny!

- **Defensive Acquisition:** The Acquirer is afraid of a high-growth competitor, so it acquires the competitor to prevent the disruption of its main business.

- **Acqui-Hire:** Recruiting top-notch employees is **expensive**. Why not shortcut the whole process by buying an entire smaller company and hiring everyone like that?

- **Office Politics, Ego, and Pride:** Let's buy this company to become bigger and better! Also, one executive can use it to win a promotion and beat out rivals for a job.

The **process** of acquiring another company is complicated and can take several months to several years to complete.

If you're an investment banker advising a company that wants to **sell**, here's a summary of the most common steps in the process:

Description:	Timing:	Responsibility:
Plan Process and Create Marketing Materials.	~1 month.	
Determine length/breadth of process.		Adviser, Seller.
Think about potential buyers.		Adviser, Seller.
Value the company and set pricing expectations.		Adviser.
Prepare "Teaser" and Offering Memorandum (OM).		Adviser, Seller.
Contact Initial Set of Potential Buyers.	~1 month.	
Send "Teaser" to buyers after contacting them.		Adviser.
Negotiate NDAs / other confidentiality agreements.		Adviser, Lawyers.
Distribute Offering Memorandum to interested parties.		Adviser.
Set up meetings with interested buyers.		Adviser, Potential Buyers.
Handle information requests and due diligence.		Adviser, Seller, Potential Buyers.
Management Meetings and Presentations.	~1 month.	
Schedule a "roadshow" with all interested buyers.		Adviser, Potential Buyers.
Prepare management team for presentations and questions.		Adviser, Seller.
Conduct presentations and answer requests and questions.		Adviser, Seller, Potential Buyers.
Initial Bids / Negotiations with Interested Buyers.	~1-2 months.	
Answer information requests from buyers.		Adviser, Seller, Potential Buyers.
Set deadline for Indications of Interest (IOIs) and receive them.		Adviser, Potential Buyers.
Evaluate and negotiate term sheets.		Adviser, Seller.
Potentially conduct additional rounds of bidding.		Adviser, Seller, Potential Buyers.
Select final buyer / buyer(s).		Adviser, Seller.
Final Negotiations, Financing, and Deal Closing.	~1-2 months.	
Conduct final due diligence.		Adviser, Seller, Lawyers, Buyer.
Negotiate Definitive Agreement.		Adviser, Seller, Lawyers, Buyer.
Arrange debt/stock financing.		Buyer, Buyer's Advisers.
Complete regulatory filings (HSR, anti-trust, etc.).		Seller, Buyer, Lawyers.
Close / announce deal.		Adviser, Seller, Lawyers, Buyer.

In a **buy-side M&A deal**, where your bank is helping one company *buy* another one, the process is similar, but it's more about *finding* companies to acquire.

Therefore, Step 1 is more about researching the market, finding the best acquisition targets, and estimating their prices.

In Step 2, you'll contact this initial set of potential Sellers, gauge their interest in selling, and collect information from them.

Steps 3 – 5 proceed as described above, but you're representing **the other party** – so you help the Buyer assess the information that Sellers send and ask follow-up questions.

And then, you help the Buyer arrange financing and negotiate the deal's final terms.

How to Finance Acquisitions

A company can fund its operations with Debt and Equity, and it can also use Debt and Equity to **acquire other companies**.

If the company has Excess Cash on its Balance Sheet, it can also use that Cash for acquisitions.

In a valuation, Debt is cheaper than Equity, and it's the same in M&A deals: Companies prefer to fund acquisitions with Debt because it's cheaper than Equity.

However, if the company has **Excess Cash**, it almost always prefers to use it first because Cash is cheaper than Debt.

Using Cash or Debt to pay for another company is straightforward: Record the Cash reduction or Debt increase and the reduced Interest Income or increased Interest Expense afterward.

Equity is a bit different because the Buyer can use two methods:

1) **It can issue new shares to OTHER investors**, collect cash from those investors, and use that cash to pay for the Seller.

2) **Or it can issue shares directly TO the Seller's shareholders** in exchange for their shares.

The financial modeling works the same way regardless of the method used: The Seller's shares disappear since it's no longer an independent entity, and the Buyer's shares increase.

These additional shares **cost** the Buyer because they **dilute** its existing investors.

For example, if a shareholder previously owned 11% of the company, it might own only 8% or 9% after the company uses Stock to fund an acquisition.

As a result, *the investor's stake in the company decreased*, even though the investor didn't sell any of their stake.

Here's a summary of the different financing methods in M&A deals:

Purchase Method:	Advantages:	Disadvantages:
Cash	Typically the cheapest method. Seller gets cash immediately. No need for time-consuming financing.	Seller also gets taxed immediately. Seller can't take advantage of potential upside in buyer's stock price.
Debt	Typically cheaper than stock (Cost of Debt vs. Cost of Equity). Seller gets cash immediately.	Increased debt profile for combined company (likely higher interest in the future and more constraints). Financing can be expensive/time-consuming. Seller gets taxed immediately. Seller can't take advantage of potential upside in buyer's stock price.
Stock	*Can* sometimes be cheaper if the buyer has a high stock price and P / E multiple. Can be faster than raising debt financing. Seller gets to participate in potential upside of buyer's stock price. Seller isn't taxed until stock is sold.	More risk for the seller since the buyer's share price could change. There may be lock-up periods for the stock and the seller might have to hold it for a long time before selling. Fixed shares vs. fixed value could make a big impact on the seller if the buyer's share price changes a lot.

Mergers vs. Acquisitions

No mechanical difference exists between a “merger” and an “acquisition.” The only difference is that the Buyer and Seller tend to be closer in size in a merger, while the Buyer tends to be much bigger in an acquisition.

If the companies are about the same size, the Buyer is unlikely to have enough Cash or Debt capacity to acquire the Seller through one of those.

So, 100% Stock or 50%+ Stock structures are far more common in mergers.

The **ownership split** is typically a major negotiating point, so the Contribution Analysis is critically important in mergers.

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Key Rule #2: Mechanics of EPS Accretion/Dilution

To assess the financial impact of an M&A deal, you create a **merger model** in Excel that:

- **Summarizes** the financial profiles of the Buyer and Seller, including projections and each company's Equity Value and Enterprise Value.
- Lists the **purchase price**, or the amount the Buyer plans to pay for the Seller, and the mix of Cash, Debt, and Stock the Buyer will use.
- Includes other **key terms** of the deal, such as the interest rates on Cash and Debt and estimates for **Synergies** – ways to boost combined revenue or reduce costs.
- Shows what the Buyer and Seller look like as a **combined entity** and how the **combined EPS** compares with the Buyer's standalone EPS.

Merger models focus on EPS because it's the only easy-to-calculate metric that captures the deal's FULL impact.

Metrics such as EBITDA and NOPAT do not capture the deal's full financial impact because they are before Interest Income and Interest Expense and do not reflect the share count.

The only other metrics that capture the **FULL** impact of the deal are Free Cash Flow per Share and Levered Free Cash Flow per Share, and these are not ideal because they are more complex to calculate, and their components vary widely in different industries.

Here's the step-by-step process for building a **simple** EPS accretion/dilution analysis, otherwise known as a merger model:

Step 1: Get the Financial Stats for the Buyer and Seller

At the minimum, you need each company's **Current Equity Value** and **Net Income**, along with everything that goes into those calculations: The Current Share Price, Shares Outstanding, Pre-Tax Income, and Tax Rate.

Here are the assumptions for our *very creatively named* "Companies A and B":

Transaction Assumptions:	Units:			Units:	
Company A Current Share Price:	$ as Stated	$ 7.00	Company B Current Share Price:	$ as Stated	$ 5.00
Company A Diluted Shares:	M Shares	100.000	Company B Diluted Shares:	M Shares	100.000
Company A Current Equity Value:	$ M	$ 700.0	Company B Current Equity Value:	$ M	$ 500.0
Company A Pre-Tax Income:	$ M	$ 100.0	Company B Pre-Tax Income:	$ M	$ 40.0
Company A Tax Rate:	%	25.0%	Company B Tax Rate:	%	25.0%
Company A Net Income:	$ M	$ 75.0	Company B Net Income:	$ M	$ 30.0

A merger model is always based on the **PROJECTED** figures in the year after the deal closes. So you must have the *projected* Revenue, Operating Income, Pre-Tax Income, Net Income, etc. for each company.

Step 2: Determine the Purchase Price and Cash/Debt/Stock Mix

In any M&A deal with a public seller, the Buyer must pay a **control premium** to do the deal.

So, if the Seller's share price is currently $5.00, the Buyer might pay $6.00 per share, representing a 20% premium.

A Buyer pays this control premium because it must **incentivize large institutional shareholders to sell their shares** – if it offered no premium, they would have no reason to sell everything.

In real life, you would also value the Seller (Company B) with a DCF and comparable companies/transactions to cross-check this premium assumption.

Here's our setup for the Equity Purchase Price, also known as the Purchase Equity Value:

	A	B	C	D	E	F	G	H	I	J	K
1											
2			**Merger Model - Illustrative Mechanics Between a Buyer and Seller**								
3			($ in Millions, Except Per Share Amounts in Dollars as Stated)								
4											
5			**Transaction Assumptions:**	*Units:*						*Units:*	
6											
7			Company A Current Share Price:	*$ as Stated*	$ 7.00		Company B Current Share Price:			*$ as Stated*	$ 5.00
8			Company A Diluted Shares:	*M Shares*	100.000		Company B Diluted Shares:			*M Shares*	100.000
9			Company A Current Equity Value:	*$ M*	$ 700.0		Company B Current Equity Value:			*$ M*	$ 500.0
10											
11			Company A Pre-Tax Income:	*$ M*	$ 100.0		Company B Pre-Tax Income:			*$ M*	$ 40.0
12			Company A Tax Rate:	*%*	25.0%		Company B Tax Rate:			*%*	25.0%
13			Company A Net Income:	*$ M*	$ 75.0		Company B Net Income:			*$ M*	$ 30.0
14											
15			Premium Paid for Company B:	*%*	20.0%		Pre-Tax Synergies:			*$ M*	
16			**Company B - Purchase Equity Value:**	*$ M*	=K9*(1+Premium_Paid)						

The **Purchase Equity Value** here is $600 million because $6.00 per share * 100 million shares = $600 million.

For private companies, the Purchase Price may be based on a multiple of EBITDA, Revenue, or another financial metric rather than a premium to the company's current share price.

Next, you estimate how much of this $600 million Purchase Equity Value can be funded with Cash, Debt, and Stock.

"But wait!" you say, "Shouldn't we use the Purchase Enterprise Value for this calculation instead? Enterprise Value is the true cost of acquiring the company!"

Yes, the Purchase Enterprise Value better represents the true acquisition price.

But in most M&A deals, the Seller's existing Debt is refinanced and replaced with the same amount of new Debt, and items like the Unfunded Pensions and Noncontrolling Interests do not change.

So, although the Buyer *pays* the Purchase Enterprise Value, the Purchase Equity Value portion makes the biggest *net impact* on the combined financial statements.

There may be additional adjustments based on the fees and the treatment of the Seller's Cash, but in simple models, the Purchase Equity Value is always the starting point.

In this example, we assume that the Seller's Cash = the Seller's Debt, so Purchase Equity Value = Purchase Enterprise Value.

Once you have the Purchase Price, you estimate the percentages of Cash, Debt, and Stock used to fund the deal.

You start by using as much of the Buyer's Cash balance as possible because it is the cheapest financing method in most cases.

Then, if you still need additional funding, you may assume a reasonable amount of Debt.

For example, maybe the Buyer has Debt of $300 million and EBITDA of $100 million for a Debt / EBITDA ratio of 3x.

Similar companies have Debt / EBITDA ratios between 4x and 5x, so it's reasonable for the Buyer to raise an additional $200 million in Debt, taking it to 5x Debt / EBITDA.

It might even be able to use more Debt if the Seller also has significant EBITDA.

If you still need more funding after that, you move to Stock.

There are no technical limits on the amount of Stock a Buyer can issue, but most companies don't want to give up majority ownership just to acquire another company.

In other cases, companies will issue Stock only up to the level at which a deal remains accretive.

We assumed a mix of 1/3 Cash, 1/3 Debt, and 1/3 Stock in this simple example.

We didn't perform the checks described above but instead assumed that these percentages represented reasonable amounts of Cash, Debt, and Stock for the Buyer.

Once you have that, you can lay out the Cash, Debt, and Stock side-by-side, along with the Cost of each method:

Premium Paid for Company B:	%	20.0%
Company B - Purchase Equity Value:	*$ M*	**$ 600.0**
% Cash:	%	33.3%
% Debt:	%	33.3%
% Stock:	%	33.3%
Company A's P / E:	x	9.3 x
Company B's P / E @ 20.0% Premium:	x	20.0 x

Transaction Financing:	Amount:	Pre-Tax Cost:	After-Tax Cost:
Cash Used:	$ 200.0	2.0%	1.5%
Debt Used:	200.0	6.0%	4.5%
Shares Issued (Millions):	28.571		10.7%
Company B Yield @ Purchase Price:			5.0%
Weighted Cost of Acquisition:			5.6%

Shares Issued is based on $600 million * 33.3% / Company A's Current Share Price.

So, if its Share Price is higher, fewer shares will be issued, and vice versa.

The Cost of Debt is like the Cost of Debt in the WACC calculation: It's the coupon rate the company would have to pay if it issued *additional* Debt today.

You could estimate it by calculating the YTM of the Buyer's Debt or peer companies' Debt or by taking the Buyer's average interest rate and assuming a slight premium (due to the added risk from the acquisition).

The Cost of Cash is based on the interest rate the company currently earns on its Cash balance, which tends to be low (close to the current Risk-Free Rate).

The Cost of Equity is based on **Buyer's Net Income / Buyer's Equity Value**, or the *reciprocal* of the Buyer's P / E multiple.

Yes, that's different from the Cost of Equity in the WACC calculation.

It's different because you are looking at the Cost of Equity in terms of *its impact on the company's EPS*, not the company's overall Discount Rate (WACC).

In most cases, this method still produces Cost of Equity figures that are higher than the After-Tax Costs of Debt and Cash.

With all the Costs set up, you can calculate the **Weighted Cost of Acquisition**:

- **Weighted Cost of Acquisition** = % Cash Used * After-Tax Cost of Cash + % Debt Used * After-Tax Cost of Debt + % Stock Used * After-Tax Cost of Stock

This Weighted Cost tells you how much the Buyer is "giving up" to acquire the Seller.

You can also calculate **Company B's Yield** at this 20% premium.

Its Yield equals its Net Income divided by the Purchase Equity Value, or $30 million / $600 million = 5.0% here.

This "Yield" is how much Company A gets in Net Income for each $1.00 it spends to acquire Company B.

You can compare the Weighted Cost of Acquisition to the Seller's Yield to determine if the deal will be accretive or dilutive.

Here's how we calculated the Weighted Cost:

16	**Company B - Purchase Equity Value:**	*$ M*	**$**	**600.0**						
17								**Pre-Tax**	**After-Tax**	
18	% Cash:	%		33.3%	**Transaction Financing:**		**Amount:**	**Cost:**	**Cost:**	
19	% Debt:	%		33.3%	Cash Used:	$	200.0	2.0%	1.5%	
20	% Stock:	%		33.3%	Debt Issued:		200.0	6.0%	4.5%	
21					Shares Issued (Millions):		28.571		10.7%	
22	Company A's P / E:	x		9.3 x	Company B Yield @ Purchase Price:				5.0%	
23	Company B's P / E @ 20.0% Premium:	x		20.0 x	Weighted Cost of Acquisition:				=Pct_Cash*K19+Pct_Debt*K20+Pct_Stock*K21	
24										
25					**The Deal Will Be:**				**Dilutive**	

EPS accretion/dilution follows these rules:

- **Weighted Cost of Acquisition < Yield of Seller:** Accretive
- **Weighted Cost of Acquisition = Yield of Seller:** Neutral
- **Weighted Cost of Acquisition > Yield of Seller:** Dilutive

In this case, the Weighted Cost of Acquisition is 5.6%, and Company B's Yield is 5.0%.

Company A pays *more* than Company B's yield, which makes the deal **dilutive**.

If the Pre-Tax Cost of Debt decreased to 3.0%, making the Weighted Cost of Acquisition 4.8%, the deal would turn accretive:

Transaction Financing:	Amount:	Pre-Tax Cost:	After-Tax Cost:
Cash Used:	$ 200.0	2.0%	1.5%
Debt Issued:	200.0	6.0%	4.5%
Shares Issued (Millions):	28.571		10.7%
Company B Yield @ Purchase Price:			5.0%
Weighted Cost of Acquisition:			5.6%
The Deal Will Be:			**Dilutive**

Transaction Financing:	Amount:	Pre-Tax Cost:	After-Tax Cost:
Cash Used:	$ 200.0	2.0%	1.5%
Debt Issued:	200.0	3.0%	2.3%
Shares Issued (Millions):	28.571		10.7%
Company B Yield @ Purchase Price:			5.0%
Weighted Cost of Acquisition:			4.8%
The Deal Will Be:			**Accretive**

There's also a special rule for **100% Stock deals**: You can compare the P / E multiples of the Buyer and Seller *at* the Seller's Purchase Equity Value to see if the deal is accretive or dilutive.

- **100% Stock Deal, Buyer's P / E > Seller's P / E at Purchase Price**: Accretive
- **100% Stock Deal, Buyer's P / E = Seller's P / E at Purchase Price**: Neutral
- **100% Stock Deal, Buyer's P / E < Seller's P / E at Purchase Price**: Dilutive

In a 100% Stock deal, the Cost of Acquisition is the reciprocal of the Buyer's P / E multiple.

So, if the Buyer's P / E is 10x, and the Seller's Purchase P / E is 5x, then the Buyer's Weighted Cost of Acquisition is 1 / 10, or 10%, and the Seller's Yield is 1 / 5, or 20%.

The Buyer pays **less** than the Seller's yield, so the deal will boost its EPS.

But if the Buyer's P / E were 10x and the Seller's Purchase P / E were 20x, the Buyer's Weighted Cost would be 1 / 10, or 10%, and the Seller's Yield would be 1 / 20, or 5%.

The Buyer pays **more** than the Seller's yield, so the deal will reduce its EPS.

% Cash:	%	0.0%
% Debt:	%	0.0%
% Stock:	%	100.0%
Company A's P / E:	x	9.3 x
Company B's P / E @ 20.0% Premium:	x	20.0 x

Transaction Financing:	Amount:	Pre-Tax Cost:	After-Tax Cost:
Cash Used:	$ -	2.0%	1.5%
Debt Issued:	-	6.0%	4.5%
Shares Issued (Millions):	85.714		10.7%
Company B Yield @ Purchase Price:			5.0%
Weighted Cost of Acquisition:			10.7%
The Deal Will Be:			**Dilutive**

It's a 100% Stock deal, and the Buyer's P / E multiple is less than the Seller's, making its Weighted Cost greater than the seller's Yield. So, the deal is Dilutive.

% Cash:	%	0.0%
% Debt:	%	0.0%
% Stock:	%	100.0%
Company A's P / E:	x	23.3 x
Company B's P / E @ 20.0% Premium:	x	20.0 x

Transaction Financing:	Amount:	Pre-Tax Cost:	After-Tax Cost:
Cash Used:	$ -	2.0%	1.5%
Debt Issued:	-	6.0%	4.5%
Shares Issued (Millions):	85.714		4.3%
Company B Yield @ Purchase Price:			5.0%
Weighted Cost of Acquisition:			4.3%
The Deal Will Be:			Accretive

It's a 100% Stock deal, and the Buyer's P / E multiple is greater than the Seller's, making its Weighted Cost less than the Seller's Yield. So, the deal is Accretive.

Step 3: Combine the Pre-Tax Incomes and Adjust for the Acquisition Effects

Once you put all the assumptions in place, you **add together** the Pre-Tax Incomes of the Buyer and Seller and factor in the **acquisition effects**:

- **Debt:** If the Buyer uses Debt, it will have to pay Interest Expense on that Debt in the future, which will reduce its Pre-Tax Income, Net Income, and EPS.
- **Stock:** If the Buyer uses Stock, it will have additional shares outstanding in the future, reducing its EPS.
- **Cash:** If the Buyer uses Cash, it will give up *future* Interest Income, reducing its Pre-Tax Income, Net Income, and EPS. This is called the "Foregone Interest on Cash."

The Foregone Interest on Cash is a true cash cost because you add the companies' projected Pre-Tax Incomes, including the *projected* Interest Income earned on Cash.

Let's say the Buyer has $180 in projected Operating Income and $20 in projected Interest Income, and the Seller has $100 in projected Pre-Tax Income.

If the Buyer uses Cash to fund the deal and earns $10 less in Interest Income as a result, you must subtract this $10 in Foregone Interest from the Combined Pre-Tax Income.

In other words, due to the Foregone Interest Income, the Combined Pre-Tax Income is $180 + $10 + $100 = $290, *not* $180 + $20 + $100 = $300!

Before you combine the Pre-Tax Incomes and factor in the acquisition effects, you can set up the analysis to show the accretion/dilution at different purchase prices:

EPS Accretion / Dilution Analysis	Units:	Premium to Paid to Company B's Share Price: 0.0%	10.0%	20.0%	30.0%	40.0%
Company A Share Price:	$ as Stated	$ 7.00	$ 7.00	$ 7.00	$ 7.00	$ 7.00
Company B Offer Price per Share:	$ as Stated	5.00	5.50	6.00	6.50	7.00
Company B - Purchase Equity Value:	$ M	**500.0**	**550.0**	**600.0**	**650.0**	**700.0**
Cash Used:	$ M	166.7	183.3	200.0	216.7	233.3
Debt Issued:	$ M	166.7	183.3	200.0	216.7	233.3
Company A Shares Issued:	M Shares	23.810	26.190	28.571	30.952	33.333
Weighted Cost of Acquisition:	%	5.6%	5.6%	5.6%	5.6%	5.6%
After-Tax Yield of Company B:	%	6.0%	5.5%	5.0%	4.6%	4.3%
The Deal is Predicted to Be:	Text	**Accretive**	**Dilutive**	**Dilutive**	**Dilutive**	**Dilutive**

The rule for predicting accretion / dilution based on the Weighted Cost of Acquisition and the Seller's Yield holds up ONLY if you keep the analysis simple and include the impact of Cash, Debt, and Stock, but nothing else.

Then, you can combine the Pre-Tax Incomes and factor in the acquisition effects:

EPS Accretion / Dilution Analysis	Units:	Premium to Paid to Company B's Share Price: 0.0%	10.0%	20.0%	30.0%	40.0%
Company A Share Price:	$ as Stated	$ 7.00	$ 7.00	$ 7.00	$ 7.00	$ 7.00
Company B Offer Price per Share:	$ as Stated	5.00	5.50	6.00	6.50	7.00
Company B - Purchase Equity Value:	$ M	**500.0**	**550.0**	**600.0**	**650.0**	**700.0**
Cash Used:	$ M	166.7	183.3	200.0	216.7	233.3
Debt Issued:	$ M	166.7	183.3	200.0	216.7	233.3
Company A Shares Issued:	M Shares	23.810	26.190	28.571	30.952	33.333
(+) Company A Pre-Tax Income:	$ M	100.0	100.0	100.0	100.0	100.0
(+) Company B Pre-Tax Income:	$ M	40.0	40.0	40.0	40.0	40.0
(-) Foregone Interest on Cash:	$ M	(3.3)	(3.7)	(4.0)	(4.3)	(4.7)
(-) Interest Paid on New Debt Issued:	$ M	(10.0)	(11.0)	(12.0)	(13.0)	(14.0)
(+) Synergies:	$ M	-	-	-	-	-
Combined Pre-Tax Income:	$ M	**126.7**	**125.3**	**124.0**	**122.7**	**121.3**

Multiply the Cash Used by the Interest Rate on Cash to get the Foregone Interest on Cash in each case.

Multiply the Debt Issued by the Interest Rate on Debt to get the Interest Paid on New Debt Issued.

You calculate "Company A Shares Issued" with (Purchase Equity Value * % Stock) / Company A's Share Price.

Step 4: Calculate the Combined Net Income and EPS

Next, you multiply the Combined Pre-Tax Income by (1 – Buyer's Tax Rate) to calculate the Combined Net Income.

Then, you add the Shares Issued in the deal to the Buyer's existing share count to get the Total Shares Outstanding.

The Seller's shares go away because it no longer exists as an independent entity – Company A has acquired all of Company B's shares, removing them from the market.

You then divide the Combined Net Income by the Total Shares Outstanding to determine the Combined EPS:

	C	D	G	H	I	J	K
43							
44	(+) Company A Pre-Tax Income:	*$ M*	100.0	100.0	100.0	100.0	100.0
45	(+) Company B Pre-Tax Income:	*$ M*	40.0	40.0	40.0	40.0	40.0
46	(-) Foregone Interest on Cash:	*$ M*	(3.3)	(3.7)	(4.0)	(4.3)	(4.7)
47	(-) Interest Paid on New Debt Issued:	*$ M*	(10.0)	(11.0)	(12.0)	(13.0)	(14.0)
48	(+) Synergies:	*$ M*	-	-	-	-	-
49	**Combined Pre-Tax Income:**	*$ M*	**126.7**	**125.3**	**124.0**	**122.7**	**121.3**
50							
51	(-) Combined Income Tax Provision:	*$ M*	=-G49*Buyer_Tax_Rate		(31.0)	(30.7)	(30.3)
52							
53	**Combined Net Income:**	*$ M*	**$ 95.0**	**$ 94.0**	**$ 93.0**	**$ 92.0**	**$ 91.0**
54							
55	(+) Company A - Diluted Shares:	*M Shares*	100.000	100.000	100.000	100.000	100.000
56	(+) Shares Issued in Transaction:	*M Shares*	23.810	26.190	28.571	30.952	33.333
57	**Total Diluted Shares:**	*M Shares*	**123.810**	**126.190**	**128.571**	**130.952**	**133.333**
58							
59	**Combined Earnings Per Share (EPS):**	*$ / Share*	**$ 0.77**	**$ 0.74**	**$ 0.72**	**$ 0.70**	**$ 0.68**

You use the **Buyer's** tax rate, even though this may not be exactly correct in real life.

You add the Shares Issued, based on the Buyer's Share Price, to the Buyer's existing share count.

In real life, this step may be more complicated; for example, the new Tax Rate might not be the Buyer's Tax Rate if you're modeling complex multinational corporations.

Also, the Buyer's Share Price often **changes** after the deal is announced, which will affect the Shares Issued in the deal.

But these assumptions are fine for case studies, modeling tests, and interview questions.

Step 5: Calculate the EPS Accretion/Dilution and Draw Conclusions

In the final step, you calculate **EPS accretion/dilution** by comparing the Combined EPS to the Buyer's Standalone EPS from before the deal took place:

52										
53	**Combined Net Income:**	*$ M*			**$ 95.0**	**$ 94.0**	**$ 93.0**	**$ 92.0**	**$ 91.0**
54									
55	(+) Company A - Diluted Shares:	*M Shares*			100.000	100.000	100.000	100.000	100.000
56	(+) Shares Issued in Transaction:	*M Shares*			23.810	26.190	28.571	30.952	33.333
57	**Total Diluted Shares:**	*M Shares*			**123.810**	**126.190**	**128.571**	**130.952**	**133.333**
58									
59	**Combined Earnings Per Share (EPS):**	*$ / Share*			**=+G53/G57**	**$ 0.74**	**$ 0.72**	**$ 0.70**	**$ 0.68**
60									
61	Company A - Standalone EPS:	*$ / Share*			$ 0.75	$ 0.75	$ 0.75	$ 0.75	$ 0.75
62									
63	Accretion / (Dilution) - $:	*$ / Share*			$ 0.02	$ (0.01)	$ (0.03)	$ (0.05)	$ (0.07)
64	Accretion / (Dilution) - %:	*%*			2.3%	(0.7%)	(3.6%)	(6.3%)	(9.0%)
65									
66	**Was Our Prediction Correct?**	*Text*			**Yes**	**Yes**	**Yes**	**Yes**	**Yes**

If the Combined EPS is higher than the Buyer's Standalone EPS, the deal is **accretive**; if the Combined EPS is lower, the deal is **dilutive**; and if it's the same, it is **neutral**.

Once you have these results, you can **use the analysis** in a few ways:

1. **Deal Screening** – You might set up simple models like the one above to screen for potential acquisitions that make sense for your client. This simple analysis doesn't tell you the whole story, but it can serve as a quick "sanity check."

2. **Pitching Ideas to Clients** – You might use this analysis to pitch potential buy-side or sell-side M&A deals to clients (e.g., "Consider a sale to this Buyer because the deal would be highly accretive for them").

3. **Deal Negotiations** – As a deal progresses, you might use the output from a merger model to push for a higher price or superior deal terms. For example, you might tell the Buyer, "Our client (the Seller) is willing to accept more in Stock and less in Cash if you raise the price by 5%; even at that level, the deal is accretive for you."

Companies never "decide" to do deals based on merger models.

They're supporting tools, like witnesses or expert testimony in a courtroom trial.

Also, dozens of factors make deal analysis more complicated for real companies.

For example, the purchase price calculations can be more complicated, **synergies** (ways to cut the combined company's costs or increase its combined revenue) can distort the EPS figures, and acquisition effects related to asset write-ups can shift the EPS numbers.

Also, EPS is not always a meaningful metric (private companies care much less about it), Net Income and "future cash flows" are *very different*, and merger models do not capture the full risks of deals or the qualitative factors required for success.

For all these reasons, real-life deal analysis goes beyond simple EPS accretion/dilution, and merger models are just the starting point.

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Key Rule #3: The Purchase Price and Cash/Debt/Stock Mix

If the Seller is public, the Buyer must pay a **premium** to its current share price or average price over the past month, quarter, or year. Otherwise, existing shareholders have no incentive to sell all their shares.

For example, let's say that Company B has a Current Share Price of $33.00, a Current Equity Value of $825 million, and a Current Enterprise Value (TEV) of $1.025 billion.

The valuation methodologies say that the company's Implied Share Price is between $35.00 and $45.00.

The premiums paid for companies in this market have been between 15% and 30% over the past ~3 years.

Therefore, an Offer Price in this range of ~$38.00 (15% premium) to ~$43.00 (30% premium) might be reasonable.

However, if the valuation methodologies said that the company's Implied Share Price were only $20.00 to $30.00, that would be irrelevant.

The Acquirer can't approach the Target and say, "Please sell to us for less than your Current Share Price – you're overvalued right now."

The Acquirer **still** must pay a premium – more than $33.00 per share – to do this deal.

Since all deals with public Sellers include control premiums, assessing the **potential Synergies** is extremely important.

If they're high enough, these Synergies + the Seller's Implied Value may still exceed the Purchase Price, justifying the deal financially.

For example, if the Seller is worth $800 million based on the valuation methodologies, but the Buyer must pay $1 billion due to the control premium, does the deal work?

It depends on how much the **Synergies** are worth. If their Present Value is higher than $200 million, the deal could work *if* it's executed properly and integration goes smoothly.

For **private companies**, the purchase price is not based on a share-price premium but on Revenue, EBITDA, or P / E multiples and the valuation methodologies.

For example, if similar companies in this market have been acquired at multiples of 8 – 12x EBITDA, a 10x TEV / EBITDA multiple might be appropriate for the Seller.

You then back into the Purchase Equity Value based on the Purchase Enterprise Value implied by this multiple.

In a simple model, it's acceptable to make the Cash, Debt, and Stock fixed percentages of the deal funding, such as 20%, 50%, and 30%.

But Acquirers almost always prefer to use *the cheapest* purchase method first, which is usually Cash.

If the Acquirer gives up only 2 – 4% interest on its Cash balance, almost any 100% Cash M&A deal looks accretive.

Once the Acquirer has used all the Cash it can, it will switch to Debt, the next cheapest acquisition method, up to a certain level, and then it will use Stock for anything past that.

The **Cash Limit** is usually based on the "Minimum Cash balance" the company needs to keep its operations running.

Depending on the Acquirer and Target's sizes, this Minimum Cash could be based on just the Acquirer's Cash balance or the Acquirer's Cash + Target's Cash.

The **Debt Limit** is usually based on a maximum Debt / EBITDA or Debt / Total Capital.

For example, maybe the combined company wants to maintain an "investment grade" credit rating, so it aims to stay below 3x Debt / EBITDA.

The **Stock Limit** is murkier because a company could potentially issue an unlimited amount of new Stock… but most companies do not want to do that.

Generally, Acquirers want to **maintain control** of the combined entity.

For example, if the Acquirer is worth $1 billion, and the Target is worth $1.2 billion, the Acquirer would not want to issue $1.2 billion of Stock to buy the Target.

It would own less than 50% of the combined entity if it did that.

Instead, it would use as much Cash and Debt as possible and switch to Stock for the remaining portion.

This limit could also be based on EPS accretion/dilution; for example, if the deal turns dilutive with over $500 million of new Stock issued, that might be the limit.

Here are a few examples of how to determine these "limits" in a simple merger model:

Merger Model - Simple Acquisition of a Public Company
($ in Millions, Except Per Share Amounts in Dollars as Stated)

Transaction Assumptions:	Units:				Units:	
Company A Current Share Price:	$ as Stated	$ 10.00		Company B Current Share Price:	$ as Stated	$ 33.00
Company A Diluted Shares:	M Shares	100.0		Company B Diluted Shares:	M Shares	25.0
Company A Tax Rate:	%	25.0%		Company B Tax Rate:	%	25.0%
Company A Equity Value:	$ M	**$ 1,000.0**		**Company B Current Equity Value:**	$ M	**$ 825.0**
(-) Cash:	$ M	(100.0)		(-) Cash:	$ M	(50.0)
(+) Debt:	$ M	400.0		(+) Debt:	$ M	250.0
Company A Enterprise Value:	$ M	**1,300.0**		**Company B Enterprise Value:**	$ M	**1,025.0**
Premium Paid for Company B:	%	20.0%		Pre-Tax Synergies:	$ M	20.0
Purchase Equity Value:	$ M	990.0		Target Minimum Cash:	$ M	30.0
Purchase Enterprise Value:	$ M	1,190.0				
Purchase TEV / Revenue:	x	1.5 x				
Purchase TEV / EBITDA:	x	9.9 x				
Purchase P / E:	x	15.5 x				
Acquirer Minimum Cash:	$ M	80.0				
Combined Cash Balance:	$ M	150.0				
Maximum Cash Available:	**$ M**	=MAX(E26-E25-K18,0)				
Maximum Combined Debt / EBITDA:	x	3.0 x				
Combined Year 1 EBITDA:	$ M	290.0				
Maximum New Debt Available:	**$ M**	**220.0**				

We add together both Cash balances to get $150 million. This $150 million is above the combined Minimum Cash of $110 million, so we can use $150 - $110 = $40 million for the deal.

Once you have the Maximum Cash Available, you move to the Maximum New Debt Available, which is based on a Maximum Combined Debt / EBITDA of 3.0x:

12	**Company A Equity Value:**	*$ M*	**$ 1,000.0**		**Company B Current Equity Value:**			*$ M*	**$ 825.0**
13	(-) Cash:	*$ M*	(100.0)		(-) Cash:			*$ M*	(50.0)
14	(+) Debt:	*$ M*	400.0		(+) Debt:			*$ M*	250.0
15	**Company A Enterprise Value:**	*$ M*	**1,300.0**		**Company B Enterprise Value:**			*$ M*	**1,025.0**
16									
17	Premium Paid for Company B:	*%*	20.0%		Pre-Tax Synergies:			*$ M*	20.0
18	Purchase Equity Value:	*$ M*	990.0		Target Minimum Cash:			*$ M*	30.0
19	Purchase Enterprise Value:	*$ M*	1,190.0						
20									
21	Purchase TEV / Revenue:	*x*	1.5 x						
22	Purchase TEV / EBITDA:	*x*	9.9 x						
23	Purchase P / E:	*x*	15.5 x						
24									
25	Acquirer Minimum Cash:	*$ M*	80.0						
26	Combined Cash Balance:	*$ M*	150.0						
27	**Maximum Cash Available:**	***$ M***	**40.0**						
28									
29	Maximum Combined Debt / EBITDA:	*x*	3.0 x						
30	Combined Year 1 EBITDA:	*$ M*	290.0						
31	**Maximum New Debt Available:**	***$ M***	=MAX(E29*E30-E14-K14,0)						
32									

The Combined Year 1 EBITDA comes from the Combined Income Statement and includes Synergies. We multiply the $290 by 3x to get $870, and then subtract the ($250 + $400) of existing Debt to get the $220 of Maximum New Debt Available.

These are just the "maximum amounts available." To determine the Cash and Debt *to use*, you use MIN functions to compare these maximums to the Purchase Equity Value:

16										
17	Premium Paid for Company B:	*%*	20.0%		Pre-Tax Synergies:			*$ M*	20.0
18	Purchase Equity Value:	*$ M*	990.0		Target Minimum Cash:			*$ M*	30.0
19	Purchase Enterprise Value:	*$ M*	1,190.0						
20							**Pre-Tax**	**After-Tax**	
21	Purchase TEV / Revenue:	*x*	1.5 x		**Transaction Financing:**	**Amount:**	**Cost:**	**Cost:**	
22	Purchase TEV / EBITDA:	*x*	9.9 x		Cash Used:	$ 40.0	2.0%	1.5%	
23	Purchase P / E:	*x*	15.5 x		New Debt Issued:	=MIN(Purchase_Equity_Value-Cash_Used,E31)			
24					Stock Issued:	730.0		7.4%	
25	Acquirer Minimum Cash:	*$ M*	80.0		Shares Issued (Millions):	73.000			
26	Combined Cash Balance:	*$ M*	150.0		Weighted Cost of Acquisition:			6.6%	
27	**Maximum Cash Available:**	***$ M***	**40.0**		Company B Yield @ Purchase Price:			6.4%	
28									
29	Maximum Combined Debt / EBITDA:	*x*	3.0 x						
30	Combined Year 1 EBITDA:	*$ M*	290.0						
31	**Maximum New Debt Available:**	***$ M***	**220.0**						
32									

Cash Used is the lesser of the Purchase Equity Value and Max Cash Available; Debt Issued is similar but subtracts the Cash Used from the Purchase Equity Value. Stock covers the remaining amount required.

You can also use this analysis to estimate the **biggest possible deal** an Acquirer could execute.

For example, let's say the Acquirer has the following profile:

- **Current Equity Value:** $500 million
- **Cash:** $200 million; **Minimum Cash:** $50 million
- **Debt:** $100 million
- **EBITDA:** $33 million
- **Debt / EBITDA:** 3.0x
- **Debt / EBITDA for Comparable Companies:** 5.0x

Without knowing *anything* about the potential Target, you can already tell a few things:

- **Cash:** The Acquirer could use a *maximum* of $150 million in Cash in any deal. This amount increases if the Target also has Excess Cash.

- **Debt:** The Acquirer could raise ~$65 million of additional Debt since $165 million of total Debt would make its Debt / EBITDA equal 5.0x. But this amount could change if the Target contributes significant EBITDA and/or has some Debt.

- **Stock:** It would be *very* unusual for this company to issue more than $500 million in Stock in any deal. Existing shareholders would lose control at that level, and the chances of dilution would increase substantially.

So, could this Acquirer purchase a Target for $400 million?

Sure – it would likely use $150 million of Cash, $65 to $100 million of Debt (depending on the acquired company's EBITDA), and Stock for $150 to $185 million.

If we had to estimate the biggest deal this Acquirer could pursue, it could use $150 million in Cash and perhaps up to $150 million in Debt, depending on Target's EBITDA.

The Acquirer's Current Equity Value is $500 million, so it would probably **not** issue more Stock than that.

So, a high-end estimate for the maximum deal size might be $150 + $150 + $500 = $800 million.

Issuing Stock worth *half* the company's Current Equity Value ($250 million) might be a more realistic maximum and would produce a maximum deal size of $550 million.

There's no scientifically proven "maximum deal size," but you can make a quick estimate using this logic.

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Key Rule #4: The "True Purchase Price" in M&A Deals

Here is the very short version:

1) **True Purchase Price** – The Purchase Enterprise Value (or something close to it) is the "true price" in an M&A deal.

2) **Seller Proceeds** – The Seller's shareholders, however, do not receive this Purchase Enterprise Value; they receive the **Purchase Equity Value** because they are only paid for their common shares.

3) **M&A Modeling** – In merger models and EPS accretion/dilution calculations, the **Purchase Equity Value** is often more relevant because the Seller's existing Cash and Debt usually make a small *net impact* on the combined statements – even though they factor into the "true price."

As a **baseline** in any M&A deal, the Buyer must pay for all the Seller's shares, corresponding to the Purchase Equity Value.

But most companies also have Cash and Debt, and most transactions have Fees and Minimum Cash requirements.

If the Seller has **Debt**, the Buyer has three options:

1) **Assume the Debt**, i.e., keep it in place as-is (rare because most debt agreements do not allow this).

2) **Refinance the Debt**, i.e., issue new Debt for the same amount and use the proceeds to repay the existing Debt (by far the most common treatment).

3) **Repay the Debt** with Cash on hand (very rare).

If the Buyer repays the Debt in full (option #3), it increases the effective price because the Buyer pays extra to do it.

This corresponds to **Enterprise Value** because you add Debt to move from Equity Value to Enterprise Value.

But even with the first two options, the Buyer *effectively* pays for the Debt because it either assumes responsibility for it or *issues new Debt* and assumes responsibility for it.

Saying that refinancing the Seller's Debt doesn't represent "payment" is like arguing that when you buy a new house via a mortgage and down payment, the new mortgage doesn't count toward the total price.

Of course it does! Yes, it's a new issuance, but the home's **total price** is still based on the equity (down payment) + debt (mortgage) used to buy it.

With the Seller's Cash, there are two main options:

1) **Distribute the Cash** to the Seller's shareholders as the deal closes (this happens in cash-free, debt-free deals for private companies).

2) **The Buyer retains the Cash** and adds it to its Balance Sheet (more common in acquisitions of public companies).

In both cases, the Seller's Cash reduces the effective acquisition price.

Again, this corresponds to **Enterprise Value** because you subtract Cash to move from Equity Value to Enterprise Value.

In the first case, the Cash distribution reduces the Purchase Equity Value, so the Buyer pays less for each of the Seller's shares.

This is because Equity Value is based on **Net Assets**, and a Cash distribution to shareholders reduces the company's Net Assets; it's like a Dividend, which also reduces Equity Value.

In the second case, the Buyer "gets" the Seller's Cash when the deal closes, which reduces its **effective price**.

Yes, maybe it paid $1,000 for the Seller's shares, but it also got the Seller's $200 Cash balance, so it only "paid" a net amount of $800 for the Seller.

Therefore, since the Seller's existing Debt always increases the effective price, and the Seller's existing Cash always reduces it, the "true price" is the Purchase Enterprise Value: Purchase Equity Value + Debt – Cash.

That said, it gets a bit more complicated because of **Fees** (for M&A advisory, legal/audit services, and financing) and **Minimum Cash** requirements.

These **increase** the effective price because they require the Buyer to pay something extra when the deal closes.

Here's an example calculation for a $990 million Purchase Equity Value deal for a Target with $250 million of Debt and $50 million of Cash:

"True Purchase Price" Calculation (Version 2):		
Purchase Equity Value:	$	990.0
(+) Target's Debt:		250.0
(-) Target's Cash:		(50.0)
(+) Transaction & Financing Fees:		24.3
(+) Target's Minimum Cash:		30.0
"True Purchase Price":	**$**	**1,244.3**

This version is closer to the amount the Buyer actually pays for the Seller and reflects the $20M of Excess Cash that can go to the Buyer.

However, in real life, no one calculates the "purchase price" this way because disclosures and data on fees and minimum Cash requirements are limited, and it's easier and simpler to use the Purchase Enterprise Value.

In a merger model, the **Purchase Equity Value** is often more relevant because the Seller's existing Cash and Debt make a small *net impact* for two main reasons:

1) **Combined Interest Income** – If the Buyer "gets" the Seller's Cash, the Combined Interest Income equals the Buyer's Interest Income + the Seller's Interest Income. The purchase price and offer premium make no difference.

If the Seller's Cash is distributed and replaced with a new Minimum Cash, the Seller's Interest Income will fall…

…but the Purchase Equity Value will also decrease, which means the Foregone Interest on Cash will be lower, offsetting this reduction in the Seller's projected Interest Income.

Therefore, the treatment of the Seller's Cash still makes a *fairly small net impact* on the Combined Income Statement in most cases.

2) **Combined Interest Expense** – If the Buyer assumes or refinances the Seller's Debt, the Combined Interest Expense equals the Buyer's Interest Expense + the Seller's Interest Expense. The deal price makes **no difference** because these items are combined. There may be a minor adjustment if the interest rate on the Debt changes.

If the Buyer repays the Seller's Debt completely, the Combined Interest Expense will change.

In this case, the Buyer's Cash balance will fall, resulting in less Interest Income, or it will issue more Stock, resulting in a higher combined Share Count.

But since the chances of a full Debt repayment are tiny, most people ignore it in models.

This Combined Income Statement in a simple merger model sums up the rationale:

Combined Income Statement:	Units:	Projected Period Year 1	Year 2	Year 3	Year 4	Year 5
Total Revenue:	*$ M*	**$ 1,800.0**	**$ 1,950.0**	**$ 2,100.0**	**$ 2,250.0**	**$ 2,400.0**
(-) Cost of Sales:	*$ M*	(1,220.0)	(1,320.0)	(1,420.0)	(1,520.0)	(1,620.0)
Gross Profit:	*$ M*	**580.0**	**630.0**	**680.0**	**730.0**	**780.0**
Operating Expenses:						
(-) SG&A:	*$ M*	(310.0)	(330.0)	(350.0)	(370.0)	(390.0)
(+) Synergies:	*$ M*	20.0	20.0	20.0	20.0	20.0
(-) Depreciation:	*$ M*	(35.0)	(42.5)	(50.0)	(57.5)	(65.0)
(-) Amortization:	*$ M*	(16.0)	(20.0)	(24.0)	(28.0)	(32.0)
Total Operating Expenses:	*$ M*	**(341.0)**	**(372.5)**	**(404.0)**	**(435.5)**	**(467.0)**
Operating Income:	*$ M*	**239.0**	**257.5**	**276.0**	**294.5**	**313.0**
(+) Interest Income:	*$ M*	3.0	3.0	3.0	3.0	3.0
(-) Foregone Interest on Cash:	*$ M*	(1.3)	(1.3)	(1.3)	(1.3)	(1.3)
(-) Interest Paid on New Debt Issued:	*$ M*	(15.4)	(15.4)	(15.4)	(15.4)	(15.4)
(-) Interest Expense:	*$ M*	(39.0)	(39.0)	(39.0)	(39.0)	(39.0)
Pre-Tax Income:	*$ M*	**186.3**	**204.8**	**223.3**	**241.8**	**260.3**
(-) Income Tax Provision:	*$ M*	(46.6)	(51.2)	(55.8)	(60.5)	(65.1)
Net Income:	*$ M*	**139.7**	**153.6**	**167.5**	**181.4**	**195.2**
(+) Acquirer - Average Shares:	*M Shares*	100.0	100.0	100.0	100.0	100.0
(+) Shares Issued in Transaction:	*M Shares*	73.0	73.0	73.0	73.0	73.0
Total Diluted Shares:	*M Shares*	**173.0**	**173.0**	**173.0**	**173.0**	**173.0**
Combined Earnings Per Share (EPS):	*$ / Share*	**$ 0.81**	**$ 0.89**	**$ 0.97**	**$ 1.05**	**$ 1.13**
Company A - Standalone EPS:	*$ / Share*	0.74	0.83	0.93	1.03	1.13
Accretion / (Dilution) - $:	*$ / Share*	0.07	0.06	0.04	0.02	0.00
Accretion / (Dilution) - %:	*%*	9.9%	6.7%	4.1%	2.0%	0.3%

We simply combine the Buyer + Seller's standalone Interest Income and Interest Expense.

In other words, the Seller's existing Cash and Debt do not *change anything* on the Combined Income Statement - only the amount the Buyer pays for the Seller's shares makes a difference vs. a simple addition of the Income Statements.

Combined Income Statement:	Units:	Projected Period Year 1	Year 2	Year 3	Year 4	Year 5
Total Revenue:	*$ M*	**$ 1,800.0**	**$ 1,950.0**	**$ 2,100.0**	**$ 2,250.0**	**$ 2,400.0**
(-) Cost of Sales:	*$ M*	(1,220.0)	(1,320.0)	(1,420.0)	(1,520.0)	(1,620.0)
Gross Profit:	*$ M*	**580.0**	**630.0**	**680.0**	**730.0**	**780.0**
Operating Expenses:						
(-) SG&A:	*$ M*	(310.0)	(330.0)	(350.0)	(370.0)	(390.0)
(+) Synergies:	*$ M*	20.0	20.0	20.0	20.0	20.0
(-) Depreciation:	*$ M*	(35.0)	(42.5)	(50.0)	(57.5)	(65.0)
(-) Amortization:	*$ M*	(16.0)	(20.0)	(24.0)	(28.0)	(32.0)
Total Operating Expenses:	*$ M*	**(341.0)**	**(372.5)**	**(404.0)**	**(435.5)**	**(467.0)**
Operating Income:	*$ M*	**239.0**	**257.5**	**276.0**	**294.5**	**313.0**
(+) Interest Income:	*$ M*	3.0	3.0	3.0	3.0	3.0
(-) Foregone Interest on Cash:	*$ M*	(1.3)	(1.3)	(1.3)	(1.3)	(1.3)
(-) Interest Paid on New Debt Issued:	*$ M*	(15.4)	(15.4)	(15.4)	(15.4)	(15.4)
(-) Interest Expense:	*$ M*	(39.0)	(39.0)	(39.0)	(39.0)	(39.0)
Pre-Tax Income:	*$ M*	**186.3**	**204.8**	**223.3**	**241.8**	**260.3**
(-) Income Tax Provision:	*$ M*	(46.6)	(51.2)	(55.8)	(60.5)	(65.1)
Net Income:	*$ M*	**139.7**	**153.6**	**167.5**	**181.4**	**195.2**
(+) Acquirer - Average Shares:	*M Shares*	100.0	100.0	100.0	100.0	100.0
(+) Shares Issued in Transaction:	*M Shares*	73.0	73.0	73.0	73.0	73.0
Total Diluted Shares:	*M Shares*	**173.0**	**173.0**	**173.0**	**173.0**	**173.0**
Combined Earnings Per Share (EPS):	*$ / Share*	**$ 0.81**	**$ 0.89**	**$ 0.97**	**$ 1.05**	**$ 1.13**
Company A - Standalone EPS:	*$ / Share*	0.74	0.83	0.93	1.03	1.13
Accretion / (Dilution) - $:	*$ / Share*	0.07	0.06	0.04	0.02	0.00
Accretion / (Dilution) - %:	*%*	9.9%	6.7%	4.1%	2.0%	0.3%

These line items are the only ones that make a difference vs. a simple addition of the Buyer + Seller Income Statements.

And they're all based on the **Purchase Equity Value**, even though it's not the "true price" in this deal.

The **bottom line** is that the Purchase Equity Value explains *most of the net changes* on the Combined Income Statement, which is why it is important despite not being the "true price."

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Key Rule #5: Cash-Free, Debt-Free Deals

Many acquisitions of private companies are structured as **cash-free, debt-free deals.**

In a cash-free debt-free deal, the Seller's existing Cash and Debt both go to $0 when the deal closes and are immediately replaced by new Debt and Cash balances. The new Cash is typically the Seller's Minimum Cash required for operations.

The **Purchase Enterprise Value** is more important in CF/DF deals, and it's always the biggest component on the Uses side of the Sources & Uses schedule.

Do NOT be fooled by the name because "cash-free, debt-free" is a slightly deceptive term.

Yes, the Seller's Cash and Debt initially go to $0, but they are *immediately replaced* and brought up to new, higher levels.

The mechanics for the Seller's Cash and Debt are as follows:

- **Debt > Cash:** The Seller uses its entire Cash balance to repay as much Debt as it can, and the remaining Debt after this repayment is deducted from the purchase price, so **the proceeds to the selling shareholders are lower**.

- **Cash > Debt:** The Target repays its entire Debt balance using its Cash, and the Target then distributes the remaining Cash to shareholders as the deal closes. **The proceeds to the selling shareholders are higher as a result**.

These mechanics mean that the selling shareholders in a cash-free, debt-free deal get the **Purchase Equity Value** because the Purchase Equity Value = Purchase Enterprise + Cash – Debt.

In ~99% of cases, cash-free, debt-free deals are **economically equivalent** to non-cash-free/debt-free deals because, in both cases, the "true price" is the Purchase Enterprise Value, the selling shareholders get the Purchase Equity Value, existing Debt is refinanced, and Cash is distributed to the shareholders and then brought up to the Minimum Cash level.

However, they *look different* at first glance, which is why many people mistakenly assume they are different.

The **deal funding** in a cash-free, debt-free deal is based on the Purchase Enterprise Value + the Seller's Minimum Cash + Transaction/Financing Fees.

So, the Sources & Uses schedule for a cash-free, debt-free deal might look like this:

Sources & Uses of Funds for Cash-Free, Debt-Free Deal:

Sources:		
Cash Used:	$	20.0
Debt Issued and Replaced:		470.0
Stock Issued:		730.0
Cash for Transaction Fees:		24.3
Total Sources:	**$**	**1,244.3**

Uses:		
Purchase Enterprise Value of Target:	$	1,190.0
Transaction Fees:		14.9
Financing Fees:		9.4
Target Minimum Cash:		30.0
Total Uses:	**$**	**1,244.3**
CHECK:		**OK!**

The **non-cash-free, debt-free version** looks like this:

Sources & Uses of Funds:

Sources:		
Cash Used:	$	40.0
New Debt Issued:		220.0
Stock Issued:		730.0
Assume/Replace Target's Debt:		250.0
Cash for Transaction Fees:		24.3
Total Sources:	**$**	**1,264.3**

Uses:		
Equity Purchase Price of Target:	$	990.0
Assume/Replace Target's Debt:		250.0
Transaction Fees:		14.9
Financing Fees:		9.4
Total Uses:	**$**	**1,264.3**
CHECK:		**OK!**

These schedules look different, but they're **equivalent** if you rearrange a few terms:

- **Purchase Enterprise Value and Existing Debt/Cash:** In the CF/DF version, the $1,190 Purchase Enterprise Value = $990 Purchase Equity Value – $50 Cash + $250 Debt. In the non-CF/DF version, the $990 Purchase Equity Value + $250 of existing Debt appear as explicit terms on the Uses side.

  The $40 of "Cash Used" on the Sources side represents $20 from the Acquirer and $20 from the Target because the Target's Cash is $50, and its Minimum Cash is $30.

  This +$20 of Cash from the Target on the Sources side is equivalent to the –$20 of Cash from the Target on the Uses side in the CF/DF version.

  That's because, in the CF/DF version, the Purchase Enterprise Value deducts the Target's $50 Cash balance and adds $30 for the Minimum Cash for a total of –$20. A negative on the Uses side is the same as a positive on the Sources side.

  The bottom line is that in both versions, the Acquirer uses $20 of Cash to fund the deal, and the Target also uses $20 of Cash to fund the deal.

- **The Debt Issued and Replaced** of $470 in the CF/DF version equals the Target's existing $250 of Debt *plus* the $220 of new Debt. In the non-CF/DF version, these are shown separately as $220 for New Debt Issued and $250 for Assume/Replace Target's Debt, but the total is still $470.

- The **Stock Issued** of $730 is the same in both versions.

On the Combined Income Statement, items such as the Foregone Interest on Cash, Interest Paid on New Debt, and New Shares Issued still exist.

There are two new line items: "Lost Interest Income" and "Interest Savings on Repaid Debt."

Again, these make the model *look* a bit different but are equivalent to the non-cash-free, debt-free version (explanation below the image):

Combined Income Statement:	Units:	Projected Period				
		Year 1	Year 2	Year 3	Year 4	Year 5
Total Revenue:	$ M	$ 1,800.0	$ 1,950.0	$ 2,100.0	$ 2,250.0	$ 2,400.0
(-) Cost of Sales:	$ M	(1,220.0)	(1,320.0)	(1,420.0)	(1,520.0)	(1,620.0)
Gross Profit:	$ M	580.0	630.0	680.0	730.0	780.0
Operating Expenses:						
(-) SG&A:	$ M	(310.0)	(330.0)	(350.0)	(370.0)	(390.0)
(+) Synergies:		20.0	20.0	20.0	20.0	20.0
(-) Depreciation:	$ M	(35.0)	(42.5)	(50.0)	(57.5)	(65.0)
(-) Amortization:		(16.0)	(20.0)	(24.0)	(28.0)	(32.0)
Total Operating Expenses:	$ M	(341.0)	(372.5)	(404.0)	(435.5)	(467.0)
Operating Income:	$ M	239.0	257.5	276.0	294.5	313.0
(+) Interest Income:	$ M	3.0	3.0	3.0	3.0	3.0
(-) Foregone Interest on Cash:	$ M	(0.9)	(0.9)	(0.9)	(0.9)	(0.9)
(-) Target's Lost Interest Income:	$ M	(0.4)	(0.4)	(0.4)	(0.4)	(0.4)
(+) Interest Savings on Repaid Debt:	$ M	15.0	15.0	15.0	15.0	15.0
(-) Interest Paid on New Debt Issued:	$ M	(32.9)	(32.9)	(32.9)	(32.9)	(32.9)
(-) Interest Expense:	$ M	(39.0)	(39.0)	(39.0)	(39.0)	(39.0)
Pre-Tax Income:	$ M	183.8	202.3	220.8	239.3	257.8
(-) Income Tax Provision:	$ M	(46.0)	(50.6)	(55.2)	(59.8)	(64.5)
Net Income:		137.9	151.7	165.6	179.5	193.4
(+) Acquirer - Average Shares:	M Shares	100.0	100.0	100.0	100.0	100.0
(+) Shares Issued in Transaction:	M Shares	73.0	73.0	73.0	73.0	73.0
Total Diluted Shares:	M Shares	173.0	173.0	173.0	173.0	173.0
Combined Earnings Per Share (EPS):	$ / Share	$ 0.80	$ 0.88	$ 0.96	$ 1.04	$ 1.12

"Lost Interest Income" is based on the Target's $20 million Excess Cash * a 2% rate.

"Interest Savings" is a simple reversal of the Target's pre-deal Interest Expense.

The Combined Income Statement is equivalent because we're just grouping items differently:

- **Foregone Interest on Cash:** The total amount was $1.3 million in the previous example. Here, it's also $0.9 + $0.4 = $1.3 million; it's just split into two lines.

- **Post-Deal Interest Expense:** The total amount was $2.5 + $15.4 + $39.0 = $56.9 million in the previous example. Here, it's $32.9 + $39.0 – $15.0 = $56.9 million.

Since the adjustments are the same, the EPS accretion/dilution is the same.

In both cases, the Target's existing Debt is replaced with new Debt at a 1% higher interest rate, the same amount of "New Debt" is used to fund the deal at the same 7% rate, and the same amount of Cash and Stock fund the deal.

Cash-free, debt-free deals are "much ado about nothing."

They look different on paper but are economically equivalent unless the treatment of Cash and Debt is wildly different, which is a "deal term" issue and not a CF/DF issue.

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Key Rule #6: How Equity Value and Enterprise Value Change in M&A Deals

Just as the financial statements change when one company acquires another, so too do metrics like Equity Value and Enterprise Value.

The rules follow from the definitions of these metrics: Enterprise Value is "capital structure-neutral" and reflects what the core business is worth, while Equity Value changes based on the capital structure and reflects the business's *net assets*, including the non-core ones.

1. **Combined Equity Value** = Acquirer's Equity Value + Market Value of Stock Issued in the Deal.

2. **Combined Enterprise Value** = Acquirer's Enterprise Value + Purchase Enterprise Value of Target.

3. **Combined Valuation Multiples:** TEV-based Combined Multiples are in between the Acquirer's current trading multiples and Target's purchase multiples. Equity Value-based Combined Multiples are usually in this range as well, *but they do not have to be*.

4. **Combined Enterprise Value-Based Multiples** – These **do not change based on the deal financing** (Cash/Stock/Debt) because the Combined Enterprise Value isn't affected by the financing method, nor are metrics like Revenue, EBIT, or EBITDA.

5. **Combined Equity Value-Based Multiples** – These **change based on the deal financing** because the Combined Equity Value changes based on the Stock issued, and the Combined Net Income changes based on the Cash and Debt used and their interest rates.

These rules do *not* necessarily hold up in the long term because companies' share prices usually change after deal announcements. The rules represent only the *immediate impact*.

We will now walk through a few simple examples of how to apply these rules using a scenario with Company A, the Acquirer, and Company B, the Target, whose financial profiles are shown below:

Company A:		Company B:	
Current Share Price:	$ 25.00	Current Share Price:	$ 20.00
Shares Outstanding:	20.0	Shares Outstanding:	5.0
Current Equity Value (Market Cap):	**$ 500.0**	**Current Equity Value (Market Cap):**	**$ 100.0**
(-) Cash & Cash-Equivalents:	(50.0)	(-) Cash & Cash-Equivalents:	(10.0)
(+) Total Debt:	100.0	(+) Total Debt:	30.0
(+) Preferred Stock:	-	(+) Preferred Stock:	-
(+) Noncontrolling Interests:	-	(+) Noncontrolling Interests:	-
Current Enterprise Value:	**550.0**	**Current Enterprise Value:**	**120.0**
Operating Income (EBIT):	48.0	Operating Income (EBIT):	21.0
(+) Depreciation & Amortization:	2.0	(+) Depreciation & Amortization:	1.0
EBITDA:	**50.0**	**EBITDA:**	**22.0**
Net Interest Expense:	(8.0)	Net Interest Expense:	(1.0)
Pre-Tax Income:	40.0	Pre-Tax Income:	20.0
Tax Rate:	25.0%	Tax Rate:	25.0%
Net Income:	**30.0**	**Net Income:**	**15.0**
Earnings per Share (EPS):	**$ 1.50**	**Earnings per Share (EPS):**	**$ 3.00**
Current Multiples:		**Current Multiples:**	
TEV / EBITDA:	11.0 x	TEV / EBITDA:	5.5 x
TEV / EBIT:	11.5 x	TEV / EBIT:	5.7 x
P / E:	16.7 x	P / E:	6.7 x

If Company A acquires Company B, the **Combined Equity Value** equals Company A's Equity Value of $550 plus the market value of Stock issued to fund the deal.

So, if it's a 100% Stock deal with no premium, Combined Equity Value = $500 + $100 = $600.

If it's a 0% Stock deal, Combined Equity Value = $500 + $0 = $500.

In this 0% Stock case, Company B's Equity Value does not "disappear" – instead, it is *transferred* into the Cash or Debt that Company A uses to acquire it.

To illustrate, here's a 100% Debt deal with no premium:

Purchase Consideration:	
Premium Paid for Company B:	0.0%
Purchase Equity Value:	$ 100.0
Purchase Enterprise Value:	120.0
% Cash:	0.0%
% Debt:	100.0%
% Stock:	0.0%
Acquirer Shares Issued:	-
Combined Share Count:	20.0
Combined Enterprise Value:	$ 670.0
Combined Equity Value:	500.0

If Company A uses no Stock in this deal, Combined Equity Value = Company A Equity Value.

If Company A pays no premium for Company B, Combined Enterprise Value = Company A Enterprise Value + Company B Enterprise Value.

With a 25% premium, a 100% Debt deal looks like this:

Purchase Consideration:	
Premium Paid for Company B:	25.0%
Purchase Equity Value:	$ 125.0
Purchase Enterprise Value:	145.0
% Cash:	0.0%
% Debt:	100.0%
% Stock:	0.0%
Acquirer Shares Issued:	-
Combined Share Count:	20.0
Combined Enterprise Value:	$ 695.0
Combined Equity Value:	500.0

With a premium paid for Company B, the only difference is that Combined Enterprise Value = Company A Enterprise Value + Company B *Purchase* Enterprise Value.

The Combined Equity Value stays the same – it's still Company A's Equity Value – because Company A doesn't issue any Stock to fund this deal.

If Company A pays a 25% premium for Company B using 50% Debt and 50% Stock, the Combined Equity Value changes, but the Combined Enterprise Value stays the same:

Purchase Consideration:	
Premium Paid for Company B:	25.0%
Purchase Equity Value:	$ 125.0
Purchase Enterprise Value:	145.0
% Cash:	0.0%
% Debt:	50.0%
% Stock:	50.0%
Acquirer Shares Issued:	2.5
Combined Share Count:	22.5
Combined Enterprise Value:	$ 695.0
Combined Equity Value:	562.5

If Stock is used in the deal, the Combined Equity Value will be higher than the Acquirer's Equity Value because you add the value of Stock issued.

But the Combined Enterprise Value stays the same!

Why Enterprise Value Stays the Same: The deal funding "replaces" the Target's old Equity Value, so all the capital is preserved.

For example, the Target's Purchase Equity Value here is $125. That goes away, but it's replaced with $62.5 of new Debt and $62.5 of new Stock.

The Combined Enterprise Value initially decreases by $125, but then it increases by $62.5 and then another $62.5, so it stays the same from beginning to end.

You can also understand this idea by returning to the definition of Enterprise Value: The value of Net Operating Assets to *all* the investors.

The Net Operating Assets of the Acquirer and Target stay the same because the financing method changes only the Cash, Debt, and Stock – so the Enterprise Value also stays the same.

Since the Combined Enterprise Value stays the same regardless of the financing method, there's another important implication:

Enterprise Value-based multiples for the Combined Company stay the same regardless of the deal financing.

Here are a few examples of this principle:

Purchase Consideration:		Purchase Consideration:		Purchase Consideration:	
Premium Paid for Company B:	25.0%	Premium Paid for Company B:	25.0%	Premium Paid for Company B:	25.0%
Purchase Equity Value:	$ 125.0	Purchase Equity Value:	$ 125.0	Purchase Equity Value:	$ 125.0
Purchase Enterprise Value:	145.0	Purchase Enterprise Value:	145.0	Purchase Enterprise Value:	145.0
% Cash:	0.0%	% Cash:	100.0%	% Cash:	33.3%
% Debt:	50.0%	% Debt:	0.0%	% Debt:	33.3%
% Stock:	50.0%	% Stock:	0.0%	% Stock:	33.3%
Acquirer Shares Issued:	2.5	Acquirer Shares Issued:	-	Acquirer Shares Issued:	1.7
Combined Share Count:	22.5	Combined Share Count:	20.0	Combined Share Count:	21.7
Combined Enterprise Value:	$ 695.0	Combined Enterprise Value:	$ 695.0	Combined Enterprise Value:	$ 695.0
Combined Equity Value:	562.5	Combined Equity Value:	500.0	Combined Equity Value:	541.7
Combined EBITDA:	72.0	Combined EBITDA:	72.0	Combined EBITDA:	72.0
Combined EBIT:	69.0	Combined EBIT:	69.0	Combined EBIT:	69.0
Combined Net Income:	42.2	Combined Net Income:	42.2	Combined Net Income:	42.2
Combined TEV / EBITDA:	9.7 x	Combined TEV / EBITDA:	9.7 x	Combined TEV / EBITDA:	9.7 x
Combined TEV / EBIT:	10.1 x	Combined TEV / EBIT:	10.1 x	Combined TEV / EBIT:	10.1 x
Combined P / E:	13.3 x	Combined P / E:	11.9 x	Combined P / E:	12.8 x

The Combined P / E multiple will change based on the Cash / Debt / Stock mix used in the deal...

But the Combined TEV / EBITDA and TEV / EBIT multiples won't change because the Combined Enterprise Value stays the same, as do the Combined EBITDA and Combined EBIT.

The Combined P / E multiple changes because the Combined Equity Value and the Combined Net Income change based on the financing method.

Stock issued affects the Combined Equity Value, and the Foregone Interest on Cash and Interest Paid on New Debt affect the Combined Net Income.

These concepts are important because many case studies, modeling tests, and interview questions are based on them.

That said, these rules are **simplifications** that ignore many real-world factors, such as synergies, additional depreciation on asset write-ups, and merger and integration costs.

Another problem is that the market may not like the deal, which usually results in the Acquirer's stock price falling.

For example, let's say that Company A announces an acquisition of Company B for $125 million, reflecting a 25% premium, but the market thinks that Company B is worth only $100 million.

Market participants can't "take away" Company A's Cash or Debt, but they can sell Company A's stock, which will reduce its stock price.

If this happens, the Purchase Equity Value and Purchase Enterprise Value will stay the same, but Company A's stock price will fall such that the Combined Equity Value and Combined Enterprise Value are both lower by $25 million.

As a result, Company A will have to issue **additional shares** to do the deal, making it less accretive.

The bottom line is that the control premium paid in an M&A deal may not last after the deal is announced.

If the market likes the deal and believes the Acquirer is paying a reasonable price for the Target, the Acquirer's share price might stay the same or even increase.

But if not, the Acquirer's share price will drop to reflect what the market believes the Target *should be worth*.

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Key Rule #7: Full Merger Model Walkthrough

In this section, we'll walk through a "full merger model" based on a real-life M&A deal: Builders FirstSource's $2.5 billion acquisition of BMC Stock Holdings.

Both companies operated in the building materials and supplies space, a mature industry at the time of the deal.

We divide this process of building a full merger model into the following eight steps:

1. Project the Financial Statements of the Buyer and Seller.
2. Estimate the Purchase Price and Deal Financing.
3. Create a Sources & Uses Schedule and Purchase Price Allocation Schedule.
4. Combine the Balance Sheets of the Buyer and Seller (OPTIONAL).
5. Calculate the Synergies (OPTIONAL).
6. Combine the Income Statements of the Buyer and Seller and Calculate Accretion / Dilution.
7. Calculate Cash Flow, Debt Repayment, and Key Metrics and Ratios.
8. Create Sensitivity Tables.

1) Project the Financial Statements of the Buyer and Seller

At the minimum, we need to project the Income Statements for the Buyer and Seller with a few key "Cash Flow" line items as well (e.g., D&A, the Change in Working Capital, and CapEx).

Here's the projected Income Statement for Builders FirstSource, the Buyer in this deal:

		Projected:				
Acquirer - Income Statement:	*Units:*	**FY20**	**FY21**	**FY22**	**FY23**	**FY24**
Revenue:						
(+) Lumber & Lumber Sheet Goods:	*$ M*	$ 2,521.8	$ 2,723.5	$ 2,941.4	$ 3,147.3	$ 3,336.1
(+) Manufactured Products:	*$ M*	1,623.5	1,753.4	1,893.6	2,026.2	2,147.8
(+) Windows, Doors & Millwork:	*$ M*	1,697.2	1,816.0	1,943.1	2,059.7	2,162.7
(+) Gypsum, Roofing & Insulation:	*$ M*	565.6	588.2	611.7	633.1	652.1
(+) Siding, Metal & Concrete Products:	*$ M*	762.5	793.0	824.8	853.6	879.2
(+) Other Building Products & Services:	*$ M*	850.8	884.9	920.3	952.5	981.0
Total Revenue:	*$ M*	**8,021.4**	**8,559.0**	**9,134.9**	**9,672.4**	**10,159.0**
(-) Cost of Sales:	*$ M*	(5,839.6)	(6,230.9)	(6,650.2)	(7,041.5)	(7,395.8)
Gross Profit:	*$ M*	**2,181.8**	**2,328.0**	**2,484.7**	**2,630.9**	**2,763.2**
Operating Expenses:						
(-) Selling, General & Administrative:	*$ M*	(1,651.6)	(1,723.5)	(1,752.4)	(1,857.5)	(1,971.7)
(-) Depreciation:	*$ M*	(96.3)	(94.1)	(91.3)	(87.1)	(91.4)
(-) Amortization of Intangibles:	*$ M*	(24.1)	(25.7)	(27.4)	(29.0)	(30.5)
Total Operating Expenses:	*$ M*	**(1,771.9)**	**(1,843.3)**	**(1,871.1)**	**(1,973.6)**	**(2,093.6)**
Operating Income:	*$ M*	**409.9**	**484.7**	**613.6**	**657.3**	**669.6**
(-) Interest Expense:	*$ M*	(104.6)	(107.1)	(87.9)	(70.6)	(58.9)
(+) Interest Income:	*$ M*	0.3	8.4	6.5	5.6	8.3
(-) Goodwill Impairment:	*$ M*	-	-	-	-	-
Pre-Tax Income:	*$ M*	**305.6**	**386.1**	**532.2**	**592.3**	**619.0**
(-) Income Tax Provision:	*$ M*	(67.2)	(84.9)	(117.1)	(130.3)	(136.2)
Net Income:	*$ M*	**238.3**	**301.2**	**415.1**	**462.0**	**482.8**
Earnings Per Share (EPS):	*$ / Share*	**$ 2.04**	**$ 2.57**	**$ 3.55**	**$ 3.95**	**$ 4.13**
Average Diluted Shares:	*M Shares*	117.0	117.0	117.0	117.0	117.0

Share count kept constant to prevent issues with EPS accretion/dilution later on.

We assumed percentage growth rates for the company's main revenue segments and made the expenses percentages of revenue or linked them to the company's employee count.

The Interest Expense is based on a weighted average coupon rate applied to the company's Debt balance, which decreases over time as the company repays Debt principal.

The Cash Flow Statement projections include recurring spending for CapEx, Acquisitions, and Changes in Debt; Stock-Based Compensation is set to $0 to keep the share count constant. Stock Issuances and Repurchases are set to $0 for the same reason.

EPS accretion/dilution is less meaningful if the pre-deal share count changes over time, so for analytical purposes, it's best to keep this count the same in the projections.

Next, we made similar projections for the Seller, which was straightforward since they were similar companies in the same industry.

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2) Estimate the Purchase Price and Deal Financing

If the Seller is **public**, the purchase price is linked to an Offer Price per Share, which should be a premium to the company's current share price.

For private Sellers, the purchase price is based on a TEV / Revenue, TEV / EBITDA, or P / E multiple, which you pick based on peer companies and the valuation methodologies.

One added complication for public companies is that the purchase price is often framed as an **Exchange Ratio**.

For example, if the Exchange Ratio is 0.5x, the Seller will receive 0.5 new shares from the Buyer for each of its current shares outstanding.

So, if the Seller currently has 100 million shares and a share price of $8.00, and the Buyer has 200 million shares and a share price of $20.00:

- The Buyer will issue 100 million * 0.5 = 50 million new shares.
- Therefore, the Buyer will pay 50 million * $20.00 = $1 billion for the Seller.
- The Offer Price per Share for the Seller is $20.00 * 0.5 = $10.00.
- $10.00 per share is 25% higher than $8.00 per share, so this is a 25% premium.

In real life, the Buyer and Seller's share prices could change significantly, so more advanced terms around the Exchange Ratio might limit the shares that can be issued.

For a basic analysis, these details are not important. Here's the first part of our assumptions:

Merger Model - Builders FirstSource, Inc. and BMC Stock Holdings
($ in Millions Except Per Share and Per Unit Data)

Affects the deal financing mix.

Transaction Assumptions:	Units:	
Transaction Close Date:	Date	2020-12-31
Valuation/Analysis Date:	Date	2020-08-26
Exchange Ratio:	x	1.3125 x
Implied Offer Price for Seller:	$ as Stated	$ 36.63
Target Share Price:	$ as Stated	$ 32.22
Premium Paid:	%	13.7%
Target - Diluted Shares @ Offer Price:	Millions	68.80
Purchase Equity Value:	$ M	$ 2,520.1
Purchase Enterprise Value:	$ M	2,608.9
Target - LTM Revenue:	$ M	$ 3,755.6
Target - LTM EBITDA:	$ M	252.8
LTM Purchase TEV / Revenue:	x	0.7 x
LTM Purchase TEV / EBITDA:	x	10.3 x

Buyer's Share Price * Exchange Ratio.

Standard calculations - TSM for Diluted Shares, but we use the Offer Price rather than Current Share Price.

	Units:	
Transaction Type: (1 = Cash/Debt/Stock; 2 = 100% Stock)	#	1
Include Revenue Synergies?	#	1
Include Expense Synergies? (1 = Yes, 0 = No)	#	1
Expense Synergy Realization Factor:	%	100.0%

These Synergy settings will factor in later on the Combined Income Statement.

The Seller's existing Debt is assumed or replaced with the same amount of new Debt (i.e., "refinanced").

Therefore, **the funds required** are based on the Purchase Equity Value, and a small amount of Cash covers the Legal, Advisory, and Debt Issuance Fees:

Target - Diluted Shares @ Offer Price:	Millions	68.80
Purchase Equity Value:	$ M	$ 2,520.1
Purchase Enterprise Value:	$ M	2,608.9
Post-Transaction Minimum Cash Balance:	$ M	$ 250.0
Post-Transaction Combined Cash Balance:	$ M	627.0
Maximum Cash Available for Deal:	$ M	377.0
Maximum Combined Debt / EBITDA:	x	4.0 x
Maximum New Debt Available for Deal:	$ M	1,195.2
New Debt - Annual Amortization:	%	1.0%
New Debt - Maturity:	# Years	5

Cash Used and Debt Used are based on MIN functions comparing the amount available to the Purchase Equity Value.

Transaction Financing:	Amount:	Pre-Tax Cost:	After-Tax Cost:
Cash Used:	$ 377.0	2.0%	1.6%
Debt Issued:	1,195.2	8.7%	6.8%
Stock Used:	947.9		9.1%
New Shares Issued (Millions):	33.963		
Weighted Average Acquisition Cost:			6.9%
Target's "Yield":			6.4%
Advisory Fee %:			1.0%
Debt Issuance Fee %:			2.0%
Legal and Other Fees:			$ 10.0

We set the Min Cash Balance a bit higher so that Cash can cover these fees.

You could simplify this by making the Cash, Debt, and Stock simple percentages, but you would need to check the dollar amounts against the restrictions here, such as the $250 million Minimum Cash post-transaction and the 4.0x maximum Debt / EBITDA.

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3) Create a Sources & Uses Schedule and Purchase Price Allocation Schedule

A **Source of Funds** is anything the Buyer uses to pay for the Seller. Cash, Debt, and Stock are the main Sources in M&A deals.

A **Use of Funds** is anything that *increases* the amount that a Buyer pays for the Seller.

The biggest Use of Funds is the Seller's Purchase Equity Value (or the Purchase Enterprise Value in cash-free, debt-free deals); deal fees and Debt repayment are other common Uses.

If the Buyer **assumes** the Seller's Debt or "refinances it" by replacing it with new Debt (what happens in most deals), that existing Debt shows up under *both* Sources and Uses:

Target - Diluted Shares @ Offer Price:	*Millions*	68.80
Purchase Equity Value:	*$ M*	$ 2,520.1
Purchase Enterprise Value:	*$ M*	2,608.9
Target - LTM Revenue:	*$ M*	$ 3,755.6
Target - LTM EBITDA:	*$ M*	252.8
LTM Purchase TEV / Revenue:	*x*	0.7 x
LTM Purchase TEV / EBITDA:	*x*	10.3 x
Post-Transaction Minimum Cash Balance:	*$ M*	$ 250.0
Post-Transaction Combined Cash Balance:	*$ M*	627.0
Maximum Cash Available for Deal:	*$ M*	377.0
Maximum Combined Debt / EBITDA:	*x*	4.0 x
Maximum New Debt Available for Deal:	*$ M*	1,195.2
New Debt - Annual Amortization:	*%*	1.0%
New Debt - Maturity:	*# Years*	5

Transaction Financing:	Amount:	Pre-Tax Cost:	After-Tax Cost:
Cash Used:	$ 377.0	2.0%	1.6%
Debt Issued:	1,195.2	8.7%	6.8%
Stock Used:	947.9		9.1%
New Shares Issued (Millions):	33.963		
Weighted Average Acquisition Cost:			6.9%
Target's "Yield":			6.4%
Advisory Fee %:			1.0%
Debt Issuance Fee %:			2.0%
Legal and Other Fees:			$ 10.0

Sources & Uses of Funds:

Sources:	
Cash Used to Purchase Equity:	$ 377.0
Debt Issued to Purchase Equity:	1,195.2
Stock Issued to Purchase Equity:	947.9
Assume/Replace Target's Debt:	336.4
Cash for Transaction Fees:	59.1
Total Sources:	**$ 2,915.6**

Uses:	
Equity Purchase Price of Target:	$ 2,520.1
Assume/Replace Target's Debt:	336.4
Transaction Fees:	35.2
Financing Fees:	23.9
Total Uses:	**$ 2,915.6**
CHECK:	**OK!**

Even though the Purchase Equity Value appears in the Sources & Uses schedule, the Purchase Enterprise Value is still the "true price" in this deal because the existing Debt that gets replaced is now the Buyer's responsibility, and the Seller's Excess Cash funds the deal.

Once you've created the Sources & Uses schedule, you move on to the **Purchase Price Allocation (PPA) schedule**.

You need to set up this schedule because, in almost all M&A deals, the Buyer pays a *premium* to the Seller's Book Value or Common Shareholders' Equity.

For example, the Buyer might pay $1,000 for the Purchase Equity Value, but the Seller's Common Shareholders' Equity is worth only $700.

This Common Shareholders' Equity is written down in the deal because the Seller no longer exists, and it's replaced with the $1,000 of Debt, Cash, or Stock used to pay for the Purchase Equity Value.

This creates a problem because you're writing down $700 on the L&E side of the Balance Sheet but replacing it with $1,000 of new items, so it goes out of balance.

To fix this problem, we create two new Assets:

1) Other Intangible Assets, which represent *identifiable* items that have some value, such as trademarks, patents, the brand name, and customer relationships; and

2) Goodwill, to represent everything else.

At the same time, the Buyer must re-value all the Seller's Assets and Liabilities when the deal closes and write them up or down to their "fair market values."

For example, perhaps the Seller purchased a building for $1,000 five years ago. This building will be listed on the Seller's Balance Sheet at $1,000 minus Accumulated Depreciation.

If the Accumulated Depreciation is $100, the building will be **$900** on the Balance Sheet.

When the Buyer reviews this building and researches the local market, it concludes that it is worth **$1,100** based on recent sales data.

Therefore, the Buyer will record a "write-up" of $200 to reflect this fair market value.

In addition to these Asset write-ups, you also write down a Seller's existing Deferred Tax Assets (DTAs) and Deferred Tax Liabilities (DTLs) in most M&A deals.

These items were created due to *temporary differences* between Cash Taxes and Book Taxes that will eventually reverse.

But these types of temporary differences are reconciled and closed out in M&A deals, so the existing DTAs and DTLs are wiped out, and a new DTL may be created if there's a write-up of PP&E or Other Intangible Assets.

Depreciation & Amortization on these items is **not** deductible for cash-tax purposes, but the company still records the D&A on its book financial statements.

Therefore, the Cash Taxes the company pays to the government will be **higher** than the Book Taxes it records on its Income Statement.

A Deferred Tax Liability means the company expects to pay **higher Cash Taxes than Book Taxes in the future**, which corresponds to this scenario.

Over time, the DTL will decrease as the company pays the higher Cash Taxes; eventually, it will reach $0 once the write-ups have been fully depreciated or amortized:

These items almost always get created or written up in M&A deals.

This DTL is based on (Intangible + PP&E Write-Ups) * Tax Rate.

Example of Deferred Tax Liabilities in M&A Deals:

Deferred Tax Liabilities mean that the company's Cash Taxes will exceed its Book Taxes in the future because of non-cash tax-deductible ite

Intangibles Created:	$ 100.0	Amortization Period:	5
PP&E Write-Up:	40.0	Depreciation Period:	8
Tax Rate:	25.0%	Initial Deferred Tax Liability:	35

	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Year 7
Book Operating Income:		$ 100	$ 110	$ 120	$ 130	$ 140	$ 150	$ 160
(-) Amortization of Intangibles:		(20)	(20)	(20)	(20)	(20)	-	-
(-) Depreciation of PP&E Write-Up:		(5)	(5)	(5)	(5)	(5)	(5)	(5)
Book Taxable Income:		**75**	**85**	**95**	**105**	**115**	**145**	**155**
Book Taxes:		19	21	24	26	29	36	39
Tax Operating Income:		100	110	120	130	140	150	160
(-) Amortization of Intangibles:		-	-	-	-	-	-	-
(-) Depreciation of PP&E Write-Up:		-	-	-	-	-	-	-
Cash Taxable Income:		**100**	**110**	**120**	**130**	**140**	**150**	**160**
Cash Taxes:		25	28	30	33	35	38	40
Deferred Tax Liability:	$ 35	$ 29	$ 23	$ 16	$ 10	$ 4	$ 3	$ 1

After an M&A deal takes place, the company's Cash Taxes will be higher than its Book Taxes because D&A from write-ups are not deductible for Cash-Tax purposes. As a result, the DTL will decrease each year until it reaches $0 when the write-ups are fully amortized or depreciated.

Here's the Purchase Price Allocation schedule for this deal:

Standard items; you always write off the Seller's Common Shareholders' Equity and create Goodwill and Other Intangible Assets to "plug the gap."

Purchase Price Allocation:

Goodwill Calculation:	
Equity Purchase Price:	**$ 2,520.1**
(-) Seller Book Value:	(1,116.1)
(+) Write-Off of Existing Goodwill:	297.1
Total Allocable Purchase Premium:	**1,701.1**
(-) Write-Up of PP&E:	(41.1)
(-) Write-Up of Intangibles:	(1,020.7)
(-) Write-Down of Deferred Tax Liabilities:	(17.7)
(+) New Deferred Tax Liability:	233.6
Total Goodwill Created:	**$ 855.3**

Fixed Asset Write-Up:	
PP&E Write-Up %:	10.0%
PP&E Write-Up Amount:	$ 41.1
Depreciation Period (Years):	8
Intangible Asset Write-Up:	
Purchase Price to Allocate:	$ 1,701.1
% Allocated to Indefinite-Lived Intangibles:	40.0%
Indefinite-Lived Intangibles:	680.5
% Allocated to Definite-Lived Intangibles:	20.0%
Intangibles Write-Up Amount:	340.2
Amortization Period (Years):	5
New Deferred Tax Liability:	$ 233.6

You write off existing DTLs or DTAs in "Stock Purchase" deals because temporary tax differences are reversed and reconciled **immediately** in an M&A deal.

The new DTL is based on the **TOTAL** amount of write-ups times the Buyer's Tax Rate. The Indefinite-Lived Intangibles are counted because they will *eventually* be written down, and that won't be cash-tax-deductible.

Asset write-ups **reduce** the Goodwill Created because they increase the Assets side of the Balance Sheet, which means less Goodwill is needed. Liability write-downs also **reduce** the Goodwill Created because they make the L&E side of the Balance Sheet smaller.

Liability write-ups **increase** the Goodwill Created because they do the opposite. Asset write-downs, similarly, **increase** the Goodwill Created.

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4) Combine the Balance Sheets of the Buyer and Seller

The Balance Sheet combination is simple if you've correctly set up the Sources & Uses and Purchase Price Allocation schedules.

You add together the Balance Sheets of the Buyer and Seller as of the deal close date, reflect the Cash, Debt, and Stock used in the deal, write up or down various Assets and Liabilities, and write down the Seller's Common Shareholders' Equity:

Balance Sheet Combination:	Units:	Acquirer FY20	Target FY20	Pro-Forma Adjustments: Debit	Credit	FY20
ASSETS:						
Current Assets:						
Cash & Cash Equivalents:	$ M	$ 422.0	$ 205.1	$ -	$ (436.1)	$ 190.9
Accounts Receivable:	$ M	771.8	350.5	-	-	1,122.3
Inventory & Other Current Assets:	$ M	668.3	445.7	-	-	1,114.0
Total Current Assets:	$ M	**1,862.0**	**1,001.3**			**2,427.2**
Long-Term Assets:						
Net Plants, Property & Equipment:	$ M	853.5	411.1	41.1	-	1,305.7
Operating Lease Assets:	$ M	325.6	152.6	-	-	478.2
Goodwill:	$ M	769.0	297.1	855.3	(297.1)	1,624.3
Other Intangible Assets:	$ M	104.3	162.0	1,020.7	-	1,287.0
Other Long-Term Assets:	$ M	22.2	8.3	-	-	30.5
Total Long-Term Assets:	$ M	**2,074.7**	**1,031.1**			**4,725.7**
TOTAL ASSETS:	$ M	**$ 3,936.7**	**$ 2,032.4**			**$ 7,152.9**

In Current Assets, you reflect the Cash used in the deal; you might also adjust the value of Inventory and adjust for Inter-Company Receivables.

In Long-Term Assets, you write up PP&E and reflect the new Goodwill and Other Intangible Assets created in the deal.

Common changes on the Liabilities & Equity side are as follows:

Balance Sheet Combination:	Units:	Acquirer FY20	Target FY20	Pro-Forma Adjustments: Debit	Credit	FY20
LIABILITIES & EQUITY:						
Current Liabilities:						
Accounts Payable:	$ M	$ 490.9	$ 176.6	-	-	667.5
Accrued Expenses & Other:	$ M	346.8	165.5	-	-	512.3
Total Current Liabilities:	$ M	**837.8**	**342.1**			**1,179.9**
Long-Term Liabilities:						
Existing Debt:	$ M	1,615.6	336.4	(336.4)	336.4	1,951.9
New Transaction Debt:	$ M	-	-	(23.9)	1,195.2	1,171.3
Operating Lease Liabilities:	$ M	331.6	159.6	-	-	491.2
Net Deferred Tax Liability (DTL):	$ M	35.0	17.7	(17.7)	233.6	268.5
Other Long-Term Liabilities:	$ M	52.2	60.5	-	-	112.8
Total Long-Term Liabilities:	$ M	**2,034.3**	**574.2**			**3,995.7**
Total Liabilities:	$ M	**$ 2,872.1**	**$ 916.3**			**$ 5,175.6**
Common Shareholders' Equity:	$ M	**$ 1,064.6**	**$ 1,116.1**	**(1,151.3)**	**947.9**	**1,977.3**
TOTAL LIABILITIES & EQUITY:	$ M	**$ 3,936.7**	**$ 2,032.4**			**$ 7,152.9**
Balance Sheet Check:		*OK!*	*OK!*			*OK!*

There are few changes in this section, but you might adjust for Inter-Company Payables.

You reflect any Debt Repaid, changes in DTLs, and New Debt here. Also, deduct the Financing Fees from the Book Value of the New Debt.

You write off the Seller's Common Shareholders' Equity, reflect the value of new Stock issued, and also deduct Legal and Advisory Fees here.

If the post-transaction Balance Sheet does **not** balance, you should review the S&U and PPA schedules and check off each item to ensure it appears on the Combined Balance Sheet.

Everything except for the Equity Purchase Price and Allocable Purchase Premium should appear once and only once in these "Pro-Forma Adjustments" columns.

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5) Calculate the Synergies

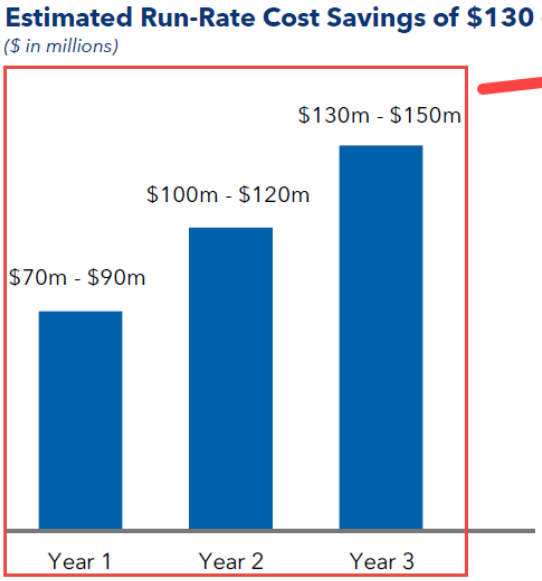
Synergies represent ways for 1 + 1 to equal 3 rather than 2 in M&A deals.

For example, the Buyer's standalone Revenue is $1,000, and the Seller's standalone Revenue is $500.

But after they merge, the Combined Revenue is $1,600 rather than $1,500 because the Buyer can sell some of its products to the Seller's customers, and vice versa.

There may also be **Expense Synergies** related to headcount reductions, building consolidation, and better supplier deals (due to volume discounts).

In simple models, you might rely on each company's estimates for Synergies and include them as lump-sum numbers:



Year 1 Expense Synergies:	$ M	$ 80.0
Year 2 Expense Synergies:	$ M	110.0
Year 3 Expense Synergies:	$ M	140.0

Combined Income Statement:	Units:	FY20	Projected - Combined Period: FY21	FY22	FY23	FY24
Total Revenue:	$ M		**12,725.2**	**13,465.6**	**14,174.5**	**14,839.7**
(-) Cost of Sales:	$ M		(9,276.5)	(9,820.3)	(10,341.6)	(10,831.4)
Gross Profit:	$ M		**3,448.8**	**3,645.3**	**3,833.0**	**4,008.3**
Operating Expenses:						
(-) Selling, General & Administrative:	$ M		(2,542.4)	(2,598.7)	(2,732.8)	(2,877.5)
(-) OpEx Synergies:	$ M		80.0	110.0	140.0	140.0
(+) Amortization of New Intangibles:	$ M		(68.0)	(68.0)	(68.0)	(68.0)
(+) Depreciation of PP&E Write-Up:	$ M		(5.1)	(5.1)	(5.1)	(5.1)
(-) Depreciation:	$ M		(156.6)	(152.0)	(150.1)	(152.3)
(-) Amortization of Existing Intangibles:	$ M		(46.5)	(49.1)	(51.5)	(49.2)
(-) Merger and Integration Costs:	$ M		(2.9)	(2.8)	(2.7)	(2.6)
(-) New Merger and Integration Costs:	$ M		(90.0)	(45.0)	-	-
(-) Impairment of Assets:	$ M		-	-	-	-
Total Operating Expenses:	$ M		**(2,831.7)**	**(2,810.7)**	**(2,870.3)**	**(3,014.8)**
Operating Income:			**617.1**	**834.6**	**962.7**	**993.5**

You can create more detailed estimates if you have more time and information.

For example, the Expense Synergies might be based on the average spending per employee and the expected headcount reduction after the deal.

It **costs money** to realize these Synergies due to employee severance costs and the need to change and consolidate internal systems, so these Merger & Integration Costs are usually modeled as a percentage of the fully realized Synergies:

Synergy Assumptions:	Units:	FY20	FY21	FY22	FY23	FY24
			Projected:			
Integration Costs % Year 3 Cost Synergies:	%	100.0%				
Total Merger & Integration Costs:	$ M	140.6				
Annual Recognition %	%		66.7%	33.3%	0.0%	0.0%
Target - Cost Synergies and Merger & Integration Costs:		**FY20**	**FY21**	**FY22**	**FY23**	**FY24**
Target - Reduction in Force:	# People		400	500	600	600
SG&A Spending per Employee:	$ as Stated		75,678	77,191	78,735	80,310
Target - Cost Savings:	$ M		30.3	38.6	47.2	48.2
Acquirer - Reduction in Force:	# People		500	750	900	900
SG&A Spending per Employee:	$ as Stated		99,677	101,670	103,704	105,778
Acquirer - Cost Savings:	$ M		49.8	76.3	93.3	95.2
Total SG&A Cost Savings:	$ M		**80.1**	**114.8**	**140.6**	**143.4**
Merger & Integration Costs:	$ M		93.7	46.9	-	-

You can then **value** these Synergies with a simple DCF; the example below also includes Revenue Synergies and subtracts the Merger & Integration Costs:

Value of Synergies to Acquirer:		FY20	FY21	FY22	FY23	FY24	FY25
After-Tax Cash Flow from Synergies:	$ M	-	3.2	78.2	147.0	162.2	163.8
Terminal Value of Synergies:	$ M						1,646.8
Present Value of Synergies:	$ M	1,218.1					
Equity Purchase Premium:	$ M	304.3					

IF we believe these Synergy figures, then this deal is a clear win for the Buyer because their PV far exceeds the premium.

This simple DCF assumes a Long-Term Growth Rate of 1% and a Discount Rate of 11% (the Buyer's WACC).

The **most common mistakes** with Synergies include "forgetting" the Merger & Integration Costs, failing to include the additional COGS and OpEx associated with Revenue Synergies, and using nonsensical numbers, such as an OpEx reduction of 50%+ in a single year.

Once you set up the Synergies and Merger & Integration Costs, you can combine the Buyer and Seller's Income Statements in the next step.

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6) Combine the Income Statements of the Buyer and Seller

Counterintuitively, we normally **start this process with the bottom section of the Cash Flow Statement** to reflect the principal repayments on the New Debt used to fund the deal.

This changing Debt balance means the Interest Expense will change each year, so we need this Debt balance *first* before combining the Income Statements.

If there are no required principal repayments, you can skip this step and return after you've finished the Combined Income Statement.

122				Projected - Combined Period:			
123	**Cash Flow and Debt Repayment:**	*Units:*	FY20	FY21	FY22	FY23	FY24
124							
145	(+) Debt Issuances:	$ M					
146	(-) Debt Repayments:	$ M					
147	(+) Issuances of Common Stock:	$ M					
148	(-) Stock Repurchases:	$ M					
149	(-) Common Dividends:	$ M					
150	(+) Issuance Fees and Other Items:	$ M					
151	(-) New Debt Principal Repayments:	$ M		=-IF((YEAR(I123)-YEAR($H123))=New_Debt_Maturity,H156,MIN(H156+SUM($H151:H151),$H$156*$E$32))			
152	**Cash Flow from Financing:**	$ M		(12.0)	(12.0)	(12.0)	(12.0)
153							
154	**Net Change in Cash:**	$ M					
155							
156	**New Debt Balance - Face Value:**	$ M	1,195.2	1,183.2	1,171.3	1,159.3	1,147.4
157	**Total Debt Balance - Face Value:**	$ M		2,905.2	2,563.3	2,321.3	2,135.4
158	**Total Cash Balance:**	$ M	190.9				

Complex-looking formula, but it just says that 100% of the Debt must be repaid in the Maturity Year; in other years, it's a fixed 1% repayment (and nothing if the Debt has already matured).

Here's each section of the Income Statement combination, starting with revenue and COGS:

		Projected - Combined Period:				
Combined Income Statement:	*Units:*	FY20	FY21	FY22	FY23	FY24
Revenue:						
(+) Lumber & Lumber Sheet Goods:	$ M		$ 3,948.6	$ 4,227.8	$ 4,498.0	$ 4,754.3
(+) Manufactured Products:	$ M		1,753.4	1,893.6	2,026.2	2,147.8
(+) Windows, Doors & Millwork:	$ M		3,041.7	3,205.6	3,360.0	3,502.0
(+) Structural Components:	$ M		748.6	786.1	825.4	866.6
(+) Gypsum, Roofing & Insulation:	$ M		588.2	611.7	633.1	652.1
(+) Siding, Metal & Concrete Products:	$ M		793.0	824.8	853.6	879.2
(+) Other Building Products & Services:	$ M		1,851.7	1,916.1	1,978.2	2,037.5
(+) Revenue Synergies:	$ M		245.1	445.4	660.2	890.3
Total Revenue:	$ M		**12,970.3**	**13,911.1**	**14,834.7**	**15,729.9**
(-) Cost of Sales:	$ M		(9,276.5)	(9,820.3)	(10,341.6)	(10,831.4)
(-) Cost of Revenue Synergies:	$ M		(179.2)	(326.1)	(483.9)	(653.5)
Total Cost of Sales:	$ M		**(9,455.6)**	**(10,146.4)**	**(10,825.5)**	**(11,484.8)**
Gross Profit:	$ M		**3,514.7**	**3,764.7**	**4,009.2**	**4,245.1**

Must include the COGS associated with Revenue Synergies as well!

With the expense combination, **be careful with the signs** because expenses should be negative, but *expense synergies* should be positive:

Combined Income Statement:	Units:	Projected - Combined Period: FY20	FY21	FY22	FY23	FY24
Operating Expenses:						
(-) Selling, General & Administrative:	$ M		(2,542.4)	(2,598.7)	(2,732.8)	(2,877.5)
(-) Operating Expenses on Revenue Synergies:	$ M		(48.2)	(87.0)	(128.4)	(172.3)
(+) OpEx Synergies:	$ M		80.1	114.8	140.6	143.4
(-) Amortization of New Intangibles:	$ M		(68.0)	(68.0)	(68.0)	(68.0)
(-) Depreciation of PP&E Write-Up:	$ M		(5.1)	(5.1)	(5.1)	(5.1)
(-) Depreciation:	$ M		(156.6)	(152.0)	(150.1)	(152.3)
(-) Amortization of Existing Intangibles:	$ M		(46.5)	(49.1)	(51.5)	(49.2)
(-) Merger and Integration Costs:	$ M		(2.9)	(2.8)	(2.7)	(2.6)
(-) New Merger and Integration Costs:	$ M		(93.7)	(46.9)	-	-
(-) Impairment of Assets:	$ M		-	-	-	-
Total Operating Expenses:	$ M		**(2,883.5)**	**(2,894.8)**	**(2,998.0)**	**(3,183.7)**
Operating Income:			**631.2**	**869.9**	**1,011.2**	**1,061.5**

Watch the signs! Expenses are negatives here, so OpEx Synergies should be positive.

Standard purchase accounting effects.

Next, we calculate the Combined Pre-Tax Income by factoring in the Net Interest Income from the Buyer and Seller, the Foregone Interest on Cash, and the Interest Paid on New Debt:

Combined Income Statement:	Units:	Projected - Combined Period: FY20	FY21	FY22	FY23	FY24
Operating Income:			**631.2**	**869.9**	**1,011.2**	**1,061.5**
(-) Interest Expense:	$ M		(128.3)	(107.4)	(88.3)	(74.6)
(+) Interest Income:	$ M		12.5	11.9	12.7	17.3
(-) Foregone Interest on Cash:	$ M		(8.7)	(8.7)	(8.7)	(8.7)
(-) Interest Paid on New Debt Issued:	$ M		(104.5)	(103.5)	(102.5)	(101.4)
(-) Amortization of New Debt Issuance Fees:	$ M		(4.8)	(4.8)	(4.8)	(4.8)
(+) Other Income, Net:	$ M		8.4	8.4	8.4	8.4
(-) Goodwill Impairment:	$ M		-	-	-	-
Pre-Tax Income:	$ M		**405.8**	**665.8**	**828.0**	**897.7**
(-) Income Tax Provision:	$ M		(89.3)	(146.5)	(182.2)	(197.5)
Net Income:	$ M		**$ 316.6**	**$ 519.3**	**$ 645.8**	**$ 700.2**
(+) Acquirer - Diluted Shares:	M Shares		117.0	117.0	117.0	117.0
(+) Shares Issued in Transaction:	M Shares		34.0	34.0	34.0	34.0
Total Diluted Shares:	M Shares		**151.0**	**151.0**	**151.0**	**151.0**
Acquirer - Standalone EPS:	$ / Share		$ 2.57	$ 3.55	$ 3.95	$ 4.13
Earnings Per Share (EPS):	$ / Share		**$ 2.10**	**$ 3.44**	**$ 4.28**	**$ 4.64**
Accretion / (Dilution) - $:	$ / Share		$ (0.48)	$ (0.11)	$ 0.33	$ 0.51
Accretion / (Dilution) - %:	%		(18.5%)	(3.0%)	8.3%	12.4%
Pro-Forma Earnings Per Share (EPS):	$ / Share		**$ 2.96**	**$ 4.06**	**$ 4.66**	**$ 5.02**
Pro-Forma Accretion / (Dilution) - $:	$ / Share		$ 0.39	$ 0.51	$ 0.71	$ 0.89
Pro-Forma Accretion / (Dilution) - %:	%		15.0%	14.4%	17.9%	21.6%

Cash Used * Interest Rate on Cash.

Changes each year as the New Debt balance decreases.

Always based on the Buyer's Tax Rate.

Based on Dollar Value of Stock Issued / Buyer's Share Price at the time of the deal.

Excludes new D&A on asset write-ups and Merger & Integration Costs.

This "Pro-Forma EPS" metric makes the deal look better by adding back the new D&A on asset write-ups and certain (allegedly) "non-recurring expenses," so many companies present it prominently when announcing deals.

It has issues, such as the lack of a standard definition for "Pro-Forma," but you should understand the basics since it is common in deal analysis.

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7) Calculate Cash Flow, Debt Repayment, and Key Metrics and Ratios

You do **not** need full Cash Flow Statements for both companies, but you do need projections for the major recurring line items, such as D&A, the Change in Working Capital, and CapEx.

We had Cash Flow Statement projections for the Buyer and Seller here, so we added together their statements and included the acquisition effects and new line items:

		Projected - Combined Period:				
Cash Flow and Debt Repayment:	*Units:*	FY20	FY21	FY22	FY23	FY24
Net Income:	*$ M*		**$ 316.6**	**$ 519.3**	**$ 645.8**	**$ 700.2**
(+) Depreciation:	*$ M*		156.6	152.0	150.1	152.3
(+) Amortization of Existing Intangibles:	*$ M*		46.5	49.1	51.5	49.2
(+) Amortization of Debt Discounts & Issuance Fees:	*$ M*		5.4	4.7	4.3	3.9
(+) Amortization of New Intangibles:	*$ M*		68.0	68.0	68.0	68.0
(+) Depreciation of PP&E Write-Up:	*$ M*		5.1	5.1	5.1	5.1
(+) Amortization of New Debt Issuance Fees:	*$ M*		4.8	4.8	4.8	4.8
(+) Goodwill Impairment:	*$ M*		-	-	-	-
(+) Stock-Based Compensation:	*$ M*		-	-	-	-
(+) Existing Deferred Income Taxes:	*$ M*		11.4	14.9	16.4	17.3
(+) New Deferred Income Taxes:	*$ M*		(16.1)	(16.1)	(16.1)	(16.1)
(+/-) Other Items:	*$ M*		0.1	0.1	0.1	0.1
(+/-) Change in Working Capital:	*$ M*		(70.3)	(73.7)	(52.5)	(44.2)
Cash Flow from Operations:	*$ M*		**528.2**	**728.2**	**877.5**	**940.7**
(-) Capital Expenditures:	*$ M*		(207.5)	(205.8)	(202.3)	(207.0)
(+) Proceeds from Sale of PP&E:	*$ M*		18.0	18.0	18.0	18.0
(-) Acquisitions:	*$ M*		(161.0)	(156.8)	(150.7)	(153.1)
Cash Flow from Investing:	*$ M*		**(350.6)**	**(344.7)**	**(335.1)**	**(342.1)**
(+) Debt Issuances:	*$ M*		975.0	1,170.0	870.0	731.9
(-) Debt Repayments:	*$ M*		(1,250.0)	(1,500.0)	(1,100.0)	(905.9)
(+) Issuances of Common Stock:	*$ M*		-	-	-	-
(-) Stock Repurchases:	*$ M*		-	-	-	-
(-) Common Dividends:	*$ M*		-	-	-	-
(+) Issuance Fees and Other Items:	*$ M*		(19.5)	(23.4)	(17.4)	(14.6)
(-) New Debt Principal Repayments:	*$ M*		(12.0)	(12.0)	(12.0)	(12.0)
Cash Flow from Financing:	*$ M*		**(306.5)**	**(365.4)**	**(259.4)**	**(200.6)**
Net Change in Cash:	*$ M*		**(128.8)**	**18.2**	**283.1**	**398.0**
New Debt Balance - Face Value:	*$ M*	**1,195.2**	**1,183.2**	**1,171.3**	**1,159.3**	**1,147.4**
Total Debt Balance - Face Value:	*$ M*		**2,905.2**	**2,563.3**	**2,321.3**	**2,135.4**
Total Cash Balance:	*$ M*	**190.9**	**62.1**	**80.3**	**363.4**	**761.4**

Direct reversals from IS line items.

Since the new D&A is not cash-tax deductible.

The Combined Cash Flow Statement is not that useful here because the New Debt always falls by 1% per year regardless of the cash flow. Optional repayments on the New Debt would make the CFS much more useful.

Once you've set up this schedule and linked everything properly, you can calculate the **Key Metrics and Ratios**.

These let you see if a deal uses too much or too little Debt and how the Combined Company differs from the Buyer and Seller as standalone entities.

This schedule is not "required," but it is helpful if you must answer case study questions or draft a presentation based on the deal:

		Projected - Combined Period:				
Combined Company - Key Metrics and Ratios:	*Units:*	**FY20**	**FY21**	**FY22**	**FY23**	**FY24**
Operating Income:	*$ M*		**$ 631.2**	**$ 869.9**	**$ 1,011.2**	**$ 1,061.5**
(+) Depreciation:	*$ M*		156.6	152.0	150.1	152.3
(+) Amortization of Existing Intangible Assets:	*$ M*		46.5	49.1	51.5	49.2
(+) Amortization of New Intangibles:	*$ M*		68.0	68.0	68.0	68.0
(+) Depreciation of PP&E Write-Up:	*$ M*		5.1	5.1	5.1	5.1
(+) Impairment Charges:	*$ M*		-	-	-	-
EBITDA:	*$ M*		**907.6**	**1,144.2**	**1,286.0**	**1,336.1**
Beginning Shareholders' Equity:	*$ M*		**1,977.3**	**2,293.9**	**2,813.2**	**3,459.1**
(+) Net Income:	*$ M*		316.6	519.3	645.8	700.2
(+) Issuances of Common Stock:	*$ M*		-	-	-	-
(-) Stock Repurchases:	*$ M*		-	-	-	-
(-) Common Dividends:	*$ M*		-	-	-	-
Ending Shareholders' Equity:	*$ M*	**1,977.3**	**2,293.9**	**2,813.2**	**3,459.1**	**4,159.3**
Revenue Growth:	*%*		*N/A*	*7.3%*	*6.6%*	*6.0%*
Operating Margin:	*%*		*4.9%*	*6.3%*	*6.8%*	*6.7%*
EBITDA Margin:	*%*		*7.0%*	*8.2%*	*8.7%*	*8.5%*
Net Margin:	*%*		*2.4%*	*3.7%*	*4.4%*	*4.5%*
Effective Tax Rate:	*%*		*22.0%*	*22.0%*	*22.0%*	*22.0%*
Depreciation % Revenue:	*%*		*1.2%*	*1.1%*	*1.0%*	*1.0%*
CapEx % Revenue:	*%*		*1.6%*	*1.5%*	*1.4%*	*1.3%*
Total Debt / EBITDA:	*x*		*3.2 x*	*2.2 x*	*1.8 x*	*1.6 x*
Net Debt / EBITDA:	*x*		*3.1 x*	*2.2 x*	*1.5 x*	*1.0 x*
EBITDA / Net Interest Expense:	*x*		*3.9 x*	*5.4 x*	*6.7 x*	*7.8 x*
Total Debt / Equity:	*x*		*1.3 x*	*0.9 x*	*0.7 x*	*0.5 x*
Total Debt / Capital:	*%*		*55.9%*	*47.7%*	*40.2%*	*33.9%*
Net Debt / Equity:	*x*		*1.2 x*	*0.9 x*	*0.6 x*	*0.3 x*
Net Debt / Net Capital:	*%*		*55.3%*	*46.9%*	*36.1%*	*24.8%*

Growth rates and margins are similar; maybe a bit higher due to Synergies.

The Combined Company de-leverages significantly, so the Buyer could probably use more Debt to fund the deal.

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8) Create Sensitivity Tables

With the model finished, we can analyze the impact of different purchase prices, different mixes of Cash, Debt, and Stock, and Synergy realization levels using **sensitivity tables**.

Here are a few examples that measure the Pro-Forma EPS accretion/dilution in Years 1 and 2:

Sensitivity Tables:

Sensitivity - Year 1 Pro-Forma EPS Accretion/Dilution vs. Exchange Ratio and Expense Synergy Realization %:

Expense Synergy Realization %:	Exchange Ratio:							
	1.1625 x	1.2125 x	1.2625 x	**1.3125 x**	1.3625 x	1.4125 x	1.4625 x	1.5125 x
160.0%	33.8%	30.6%	27.5%	**24.6%**	21.8%	19.2%	16.6%	14.2%
140.0%	30.3%	27.2%	24.2%	**21.4%**	18.7%	16.1%	13.6%	11.2%
120.0%	26.9%	23.9%	21.0%	**18.2%**	15.5%	13.0%	10.6%	8.3%
100.0%	**23.4%**	**20.5%**	**17.7%**	**15.0%**	**12.4%**	**9.9%**	**7.6%**	**5.3%**
80.0%	20.0%	17.1%	14.4%	**11.8%**	9.3%	6.9%	4.6%	2.4%
60.0%	16.5%	13.7%	11.1%	**8.5%**	6.1%	3.8%	1.6%	(0.6%)
40.0%	13.1%	10.4%	7.8%	**5.3%**	3.0%	0.7%	(1.4%)	(3.5%)
20.0%	9.6%	7.0%	4.5%	**2.1%**	(0.2%)	(2.4%)	(4.5%)	(6.5%)
0.0%	6.2%	3.6%	1.2%	**(1.1%)**	(3.3%)	(5.4%)	(7.5%)	(9.4%)

Sensitivity - Year 2 Pro-Forma EPS Accretion/Dilution vs. Exchange Ratio and Expense Synergy Realization %:

Expense Synergy Realization %:	Exchange Ratio:							
	1.1625 x	1.2125 x	1.2625 x	**1.3125 x**	1.3625 x	1.4125 x	1.4625 x	1.5125 x
160.0%	33.6%	30.4%	27.4%	**24.5%**	21.7%	19.0%	16.5%	14.0%
140.0%	30.1%	26.9%	24.0%	**21.1%**	18.4%	15.8%	13.4%	11.0%
120.0%	26.5%	23.4%	20.5%	**17.8%**	15.2%	12.6%	10.2%	7.9%
100.0%	**22.9%**	**19.9%**	**17.1%**	**14.4%**	**11.9%**	**9.4%**	**7.1%**	**4.9%**
80.0%	19.3%	16.4%	13.7%	**11.1%**	8.6%	6.2%	4.0%	1.8%
60.0%	15.7%	12.9%	10.3%	**7.8%**	5.3%	3.0%	0.8%	(1.3%)
40.0%	12.1%	9.4%	6.9%	**4.4%**	2.1%	(0.2%)	(2.3%)	(4.3%)
20.0%	8.5%	5.9%	3.4%	**1.1%**	(1.2%)	(3.4%)	(5.4%)	(7.4%)
0.0%	4.9%	2.4%	0.0%	**(2.3%)**	(4.5%)	(6.6%)	(8.6%)	(10.5%)

Deal price would have to increase significantly and Synergies would have to be a complete failure for this transaction to turn dilutive, at least on a Pro-Forma basis.

These tables are not that useful because the main question is not the proper Exchange Ratio or purchase price but **the financing**: 100% Stock vs. a mix of Cash, Debt, and Stock.

Also, we're not sensitizing **the Buyer's Share Price** because we've simplified the Exchange Ratio mechanics in this model.

We could address the first issue by setting the financing to 100% Stock and recalculating the sensitivity tables; the second issue is more difficult to analyze without changing the model significantly and making the Exchange Ratio mechanics more complex.

Remember that just because an M&A deal is accretive doesn't mean it's a good idea.

Accretion/dilution analysis is a useful screening tool and can be a starting point for deal negotiations, but EPS accretion itself does not make a deal "work."

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Key Rule #8: M&A Valuation, Value Creation, and the Contribution Analysis

There are many ways to evaluate a merger or acquisition, and EPS accretion/dilution analysis isn't the best choice in all scenarios.

Accretion/dilution makes the most sense when the Buyer and Seller are close in size, have positive EPS and cash flow, and have **financial motivations** for the deal.

But not all Buyers care about EPS, and many deals feature companies of very different sizes or with negative EPS figures.

Also, even *if* the Buyer and Seller are both similarly sized public companies with positive EPS, additional analyses may be useful for **cross-checking** your work.

If a deal uses > 50% Stock, factors such as the **post-deal ownership** and its **potential impact on the Buyer's share price** also matter a lot – which is why methods like the Contribution Analysis and the Value Creation Analysis are quite important.

Here are some alternative methods for evaluating M&A deals:

Qualitative / "Strategic" Analysis

This method is relevant in many acquisitions of small tech and biotech startups.

These small companies have no profits, no cash flow, and almost no revenue, so acquisitions are not based on financial criteria such as valuation multiples or EPS accretion/dilution.

Instead, the rationale comes down to:

1) **Potential for Extremely High Growth** – For example, if Small Biotech Firm B discovers a cure for skin cancer, Big Pharma Company A might become the most valuable healthcare company in the world if it acquires this smaller firm.

2) **Defensive Acquisition / Fear of Competition** – If Big Social Media Company A perceives a new, high-growth social media company as a threat, it might acquire it because of its strong **network effects**. If it doesn't act quickly, the smaller company might grow virally and disrupt the bigger company's entire business!

These deals have little in-depth financial analysis, but you might justify them with back-of-the-envelope math.

For example, maybe each user of Smaller Social Media Company is worth $1.00 per year, and it has 100 million users currently.

But its users could grow to 500 million in a few years, so Big Social Media Company A offers $300 million to acquire it – more than what the company is currently worth but less than what it *might be worth* in the future.

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M&A Valuation

You can value a company as a standalone entity or in the context of an M&A transaction.

In the context of an M&A transaction, the valuation should include **potential Synergies**, and **the Discount Rate** may be the Buyer's Rate or a blended Buyer/Seller Rate.

If you complete an M&A valuation, you can compare the company's **Implied Equity Value** to the **Purchase Equity Value** to evaluate whether the Buyer is paying an appropriate price (it's also fine to compare the Enterprise Values).

With the example deal used in this guide, we **valued the Seller on a standalone basis**, added the **Present Value of the Synergies**, and compared that total to the Purchase Equity Value.

In other words, does the Seller's Implied Equity Value + PV of Synergies exceed the Purchase Equity Value the Buyer is paying?

If so, the deal makes sense; if not, the Synergies must be higher for the math to work.

The UFCF projections in the standalone DCF look like this:

Unlevered Free Cash Flow Projections:	Units:	FY20	FY21	FY22	FY23	FY24	FY25	FY26	FY27	FY28	FY29
						Projected:					
Total Revenue:	$ M	$ 3,941.6	$ 4,166.2	$ 4,330.7	$ 4,502.1	$ 4,680.7	$ 4,867.9	$ 5,062.6	$ 5,265.1	$ 5,449.4	$ 5,640.1
Growth Rate:	%	8.7%	5.7%	3.9%	4.0%	4.0%	4.0%	4.0%	4.0%	3.5%	3.5%
Earnings Before Interest & Taxes (EBIT):	$ M	185.1	215.5	229.2	238.6	257.1	267.7	278.4	289.6	299.7	310.2
EBIT Margin:	%	4.7%	5.2%	5.3%	5.3%	5.5%	5.5%	5.5%	5.5%	5.5%	5.5%
(-) Taxes, Excluding Effect of Interest:	$ M	(40.7)	(47.4)	(50.4)	(52.5)	(56.6)	(58.9)	(61.3)	(63.7)	(65.9)	(68.2)
Net Operating Profit After Tax (NOPAT):	$ M	**144.4**	**168.1**	**178.8**	**186.1**	**200.6**	**208.8**	**217.2**	**225.9**	**233.8**	**242.0**
Adjustments for Non-Cash Charges:											
(+) Depreciation & Amortization:	$ M	82.8	83.3	82.3	85.5	79.6	82.8	81.0	79.0	81.7	84.6
% Revenue:	%	2.1%	2.0%	1.9%	1.9%	1.7%	1.7%	1.6%	1.5%	1.5%	1.5%
(+/-) Deferred Income Taxes:	$ M	2.6	3.1	3.3	3.4	3.6	3.8	4.0	4.1	4.3	4.4
% Book Taxes:	%	6.5%	6.5%	6.5%	6.5%	6.5%	6.5%	6.5%	6.5%	6.5%	6.5%
(+/-) Net Change in Working Capital:	$ M	(33.3)	(30.4)	(20.6)	(21.4)	(22.2)	(18.7)	(19.5)	(20.3)	(18.4)	(19.1)
% Change in Revenue:	%	(10.6%)	(13.5%)	(12.5%)	(12.5%)	(12.4%)	(10.0%)	(10.0%)	(10.0%)	(10.0%)	(10.0%)
(-) Capital Expenditures & Acquisitions:	$ M	(138.0)	(137.5)	(134.3)	(130.6)	(126.4)	(116.8)	(111.4)	(105.3)	(109.0)	(112.8)
% Revenue:	%	3.5%	3.3%	3.1%	2.9%	2.7%	2.4%	2.2%	2.0%	2.0%	2.0%
Annual Unlevered Free Cash Flow:	$ M	**58.5**	**86.6**	**109.5**	**123.1**	**135.2**	**159.8**	**171.3**	**183.4**	**192.4**	**199.1**
Growth Rate:	%	*100.2%*	*48.2%*	*26.4%*	*12.4%*	*9.8%*	*18.2%*	*7.2%*	*7.1%*	*4.9%*	*3.5%*
PV of Unlevered FCF:	$ M		79.8	92.9	96.2	97.3	106.0	104.6	103.1	99.6	95.0
EBITDA:	$ M	$ 267.8	$ 298.9	$ 311.5	$ 324.1	$ 336.7	$ 350.5	$ 359.4	$ 368.6	$ 381.5	$ 394.8
Growth Rate:	%	*16.4%*	*11.6%*	*4.2%*	*4.0%*	*3.9%*	*4.1%*	*2.6%*	*2.5%*	*3.5%*	*3.5%*

The deal closes at the end of FY 20, so FY 21 is the first "Projected Year" in the analysis.

The Seller's "undisturbed" share price just before this deal was announced was $32.22.

The DCF produces an Implied Share Price of $29.85, with a 25th to 75th percentile range of approximately $25.00 to $35.00.

So, we conclude that the Seller is *slightly overvalued* as a standalone entity.

Therefore, if the Buyer pays a **control premium** to do this deal, it must realize substantial Synergies for the math to work – the Seller is already slightly overvalued, and the Buyer is paying a 10% to 30% premium on top of that.

We previously valued the Synergies using the Acquirer's WACC of 11% and found they were worth approximately $1.2 billion. At the Seller's slightly lower WACC, they are worth closer to $1.7 billion.

If we compromise and say the Synergies are worth about $1.5 billion:

- **Seller's Implied Standalone Equity Value from DCF:** $2.0 billion
- **Present Value of Synergies:** $1.5 billion
- **Equity Purchase Price:** $2.5 billion

Since the Implied Standalone Equity Value + PV of Synergies is much higher than the Equity Purchase Price, this deal seems like a clear win from a valuation perspective.

However, this assumes that 100% of the Revenue and Expense Synergies are realized without any cost overruns or delays and that they continue indefinitely into the future.

If the Synergies are worth only **50%** of our expectations ($750 million), the purchase price and valuation are much closer to breakeven.

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IRR vs. Discount Rate (WACC)

If you've already set up a DCF to value the Seller, you've projected and valued the possible Synergies, and you've calculated WACC for the Buyer and Seller, you can easily add an **IRR vs. Discount Rate** analysis.

The Discount Rate represents the Buyer's **targeted annualized returns**, and the IRR represents **the annualized returns the Buyer might earn on the deal.**

Therefore, if the IRR exceeds the Discount Rate, the deal is good; if not, the deal is not.

The mechanics are simple, but the analysis requires several judgment calls:

1) **Which Discount Rate Do You Use?** Probably not the Seller's, but do you use the Buyer's Discount Rate or a "weighted average"?

2) **Do You Use UFCF and WACC or LFCF and Cost of Equity?** We prefer UFCF and WACC because the DCF already uses UFCF and WACC, and Levered DCFs tend to create consistency issues.

3) **To Whom Do You Attribute the Synergies?** Does the Buyer get all the credit? Or is the credit split proportionally between the Seller and Buyer?

With the first question, we use **the Buyer's Discount Rate** as the baseline in this deal but sensitize it to include numbers closer to the Seller's WACC as well.

With the third question, we effectively attribute the Synergies to the **Seller's cash flows** because the Buyer would not be able to realize any of these without the Seller.

Here's our analysis, which extends directly from the DCF:

73												
74	**Annual Unlevered Free Cash Flow:**	*$ M*	**58.5**	**86.6**	**109.5**	**123.1**	**135.2**	**159.8**	**171.3**	**183.4**	**192.4**	**199.1**
75	*Growth Rate:*	*%*	*100.2%*	*48.2%*	*26.4%*	*12.4%*	*9.8%*	*18.2%*	*7.2%*	*7.1%*	*4.9%*	*3.5%*
76												
77	PV of Unlevered FCF:	*$ M*	58.5	79.8	92.9	96.2	97.3	106.0	104.6	103.1	99.6	95.0
78												
79	**EBITDA:**	*$ M*	**$ 267.8**	**$ 298.9**	**$ 311.5**	**$ 324.1**	**$ 336.7**	**$ 350.5**	**$ 359.4**	**$ 368.6**	**$ 381.5**	**$ 394.8**
80	*Growth Rate:*	*%*	*16.4%*	*11.6%*	*4.2%*	*4.0%*	*3.9%*	*4.1%*	*2.6%*	*2.5%*	*3.5%*	*3.5%*
81												
82	**Sensitivity Tables:**											
111												
112	**IRR vs. Discount Rate:**	***Units:***	**FY20**	**FY21**	**FY22**	**FY23**	**FY24**	**FY25**	**FY26**	**FY27**	**FY28**	**FY29**
113												
114	Purchase Enterprise Value:	*$ M*	$ 2,608.9									
115	Acquirer's WACC:	%	11.0%									
116	Long-Term Growth Rate of Synergies:	%	1.0%									
117	Synergy Realization Factor:	%	100.0%									
118												
119	Unlevered Free Cash Flow of Target:	*$ M*		=I74	109.5	123.1	135.2	159.8	171.3	183.4	192.4	199.1
120	After-Tax Cash Flow from Synergies:	*$ M*		3.2	78.2	147.0	162.2	163.8	165.4	167.1	168.8	170.4
121	Terminal Value of Target UFCF:	*$ M*										2,001.6
122	Terminal Value of Synergies:	*$ M*										1,713.6
123	**Total Unlevered FCF from Deal:**	*$ M*	**(2,608.9)**	**89.9**	**187.8**	**270.1**	**297.4**	**323.6**	**336.7**	**350.5**	**361.1**	**4,084.8**
124												
125	**Internal Rate of Return (IRR):**	%	**12.8%**									

The baseline Unlevered IRR of ~13% exceeds the Acquirer's WACC of 11%, and sensitivity tables confirm this continues even at higher purchase prices.

However, the Synergies present a significant risk because ~80%+ of the Expected Synergies must be realized for the IRR to exceed the Discount Rate reliably.

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Relative Contribution Analysis

Another way to assess mergers and acquisitions is the **Relative Contribution Analysis**.

In other words, if the Acquirer contributes 80% of the Combined Company's Revenue, EBITDA, and other financial metrics, will it *own* 80% of the Combined Company afterward?

If it owns *less than* 80%, then perhaps it is paying *too much* for the Target; if it owns *more than* 80%, then perhaps it is paying *too little* for the Target.

The Contribution Analysis is **most relevant** for 100% Stock deals, "Merger of Equal" (MOE) deals, private company M&A deals, and deals financed with at least 50% Stock.

It's not relevant for 100% Cash or 100% Debt deals because the Target won't have any ownership in the Combined Company.

You start this analysis by summing up the relevant metrics for the Acquirer and Target and calculating the Contribution Percentage for each one:

Relative Contribution Analysis:	Contribution ($):			Contribution (%):	
Equity Value:	**Acquirer:**	**Target:**	**Pro-Forma:**	**Acquirer:**	**Target:**
Undisturbed Equity Value:	$ 3,323.6	$ 2,215.8	$ 5,539.4	60.0%	40.0%
Pro-Forma Ownership @ 13.7% Premium:	3,323.6	2,520.1	5,843.7	56.9%	43.1%
Revenue:					
Combined Year 1:	8,559.0	4,166.2	12,725.2	67.3%	32.7%
Combined Year 2:	9,134.9	4,330.7	13,465.6	67.8%	32.2%
EBIT:					
Combined Year 1:	484.7	215.5	700.3	69.2%	30.8%
Combined Year 2:	613.6	229.2	842.8	72.8%	27.2%
EBITDA:					
Combined Year 1:	604.5	298.9	903.4	66.9%	33.1%
Combined Year 2:	732.3	311.5	1,043.8	70.2%	29.8%
Net Income:					
Combined Year 1:	301.2	161.3	462.5	65.1%	34.9%
Combined Year 2:	415.1	174.3	589.4	70.4%	29.6%
Balance Sheet Items as of Transaction Close Date (Ignores Acquisition Effects):					
Total Assets:	3,936.7	2,032.4	5,969.1	66.0%	34.0%
Common Shareholders' Equity:	1,064.6	1,116.1	2,180.8	48.8%	51.2%

Then, you calculate each metric's Combined Pro-Forma Enterprise Value based on the Acquirer's Current Enterprise Value and its Contribution Percentage.

So, if the Acquirer's Current Enterprise Value is $3 billion, and it contributes 75% of the Revenue, the Combined Pro-Forma Enterprise Value is $3 billion / 75%, or $4 billion.

That $4 billion represents what the Combined Enterprise Value "should be" for this new company once the deal closes.

For Equity Value-based metrics, you use the Acquirer's Current Equity Value and Contribution Percentage to determine the Combined Pro-Forma Equity Value.

To move from the Combined Pro-Forma Enterprise Value to the Combined Pro-Forma Equity Value in each case, you subtract or add the standard TEV "bridge items" *for both companies*.

Then, you calculate the Target's Implied Value based on *each metric*.

For TEV-based metrics, take the Combined Pro-Forma Enterprise Value and subtract the Acquirer's Current Enterprise Value. Here's an example for the Combined Revenue metrics:

Target - Equity Value and Enterprise Value:	
Purchase Equity Value:	**$ 2,520.1**
(-) Cash & Cash-Equivalents:	(253.4)
(-) Equity Investments & Non-Core Assets:	(18.0)
(+) Total Debt & Finance Leases:	360.2
(+) Preferred Stock:	-
(+) Noncontrolling Interests & Other:	-
Purchase Enterprise Value:	**$ 2,608.9**

Acquirer - Equity Value and Enterprise Value:	
Current Equity Value:	**$ 3,323.6**
(-) Cash & Cash-Equivalents:	(385.5)
(-) Equity Investments & Non-Core Assets:	(13.6)
(+) Total Debt & Finance Leases:	1,634.5
(+) Preferred Stock:	-
(+) Noncontrolling Interests & Other:	-
Current Enterprise Value:	**$ 4,559.1**

Relative Contribution Analysis:	Contribution ($):			Contribution (%):		Combined Pro-Forma Value:		Implied Target Value:	
Equity Value:	**Acquirer:**	**Target:**	**Pro-Forma:**	**Acquirer:**	**Target:**	**Enterprise Value:**	**Equity Value:**	**Enterprise Value:**	**Equity Value:**
Undisturbed Equity Value:	$ 3,323.6	$ 2,215.8	$ 5,539.4	60.0%	40.0%	$ 6,863.7	$ 5,539.4	$ 2,304.6	$ 2,215.8
Pro-Forma Ownership @ 13.7% Premium:	3,323.6	2,520.1	5,843.7	56.9%	43.1%	7,168.0	5,843.7	2,608.9	2,520.1
Revenue:									
Combined Year 1:	8,559.0	4,166.2	12,725.2	67.3%	32.7%	6,778.3	5,454.0	=I23-J13	2,130.4
Combined Year 2:	9,134.9	4,330.7	13,465.6	67.8%	32.2%	6,720.5	5,396.2	2,161.4	2,072.6

You then move to the Target's Implied Equity Value by subtracting the items in the Equity Value to Enterprise Value bridge.

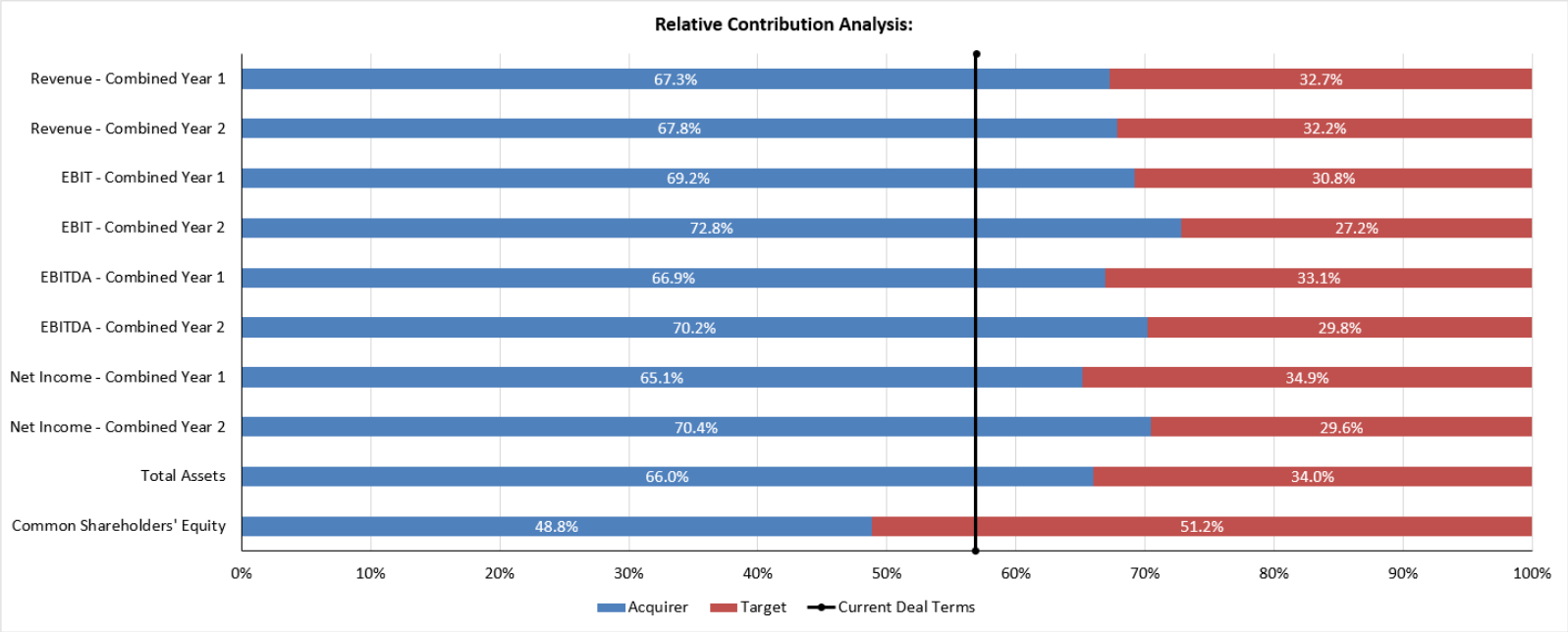
These numbers tell you what the Equity Purchase Price "should be," based on the percentage the Target contributes to each combined financial metric.

You can also divide by the Target's share count to get the Purchase Price per Share, and you can divide that by the Acquirer's current share price to get the Implied Exchange Ratio, as shown on the right.

Most of these purchase prices are below the Offer Price per Share of $36.63, so the Buyer may be overpaying.

Relative Contribution Analysis:	Implied Figures in 100% Stock Deal:		
Equity Value:	**Exchange Ratio:**	**Purchase Price:**	**Variance to Offer Price:**
Undisturbed Equity Value:	1.1544 x	$ 32.22	(12.0%)
Pro-Forma Ownership @ 13.7% Premium:	1.3130 x	36.65	0.0%
Revenue:			
Combined Year 1:	1.1099 x	30.98	(15.4%)
Combined Year 2:	1.0798 x	30.14	(17.7%)
EBIT:			
Combined Year 1:	1.0099 x	28.19	(23.1%)
Combined Year 2:	0.8411 x	23.47	(35.9%)
EBITDA:			
Combined Year 1:	1.1280 x	31.48	(14.1%)
Combined Year 2:	0.9641 x	26.91	(26.5%)
Net Income:			
Combined Year 1:	0.9275 x	25.89	(29.3%)
Combined Year 2:	0.7271 x	20.29	(44.6%)
Balance Sheet Items as of Transaction Close Date (Ignores Acquisition Effects):			
Total Assets:	0.8940 x	24.95	(31.9%)
Common Shareholders' Equity:	1.8153 x	50.67	38.3%

You can also create a graph showing the **Contribution Percentage** for each metric vs. the Ownership Percentage in a 100% Stock verison of this deal:



So, based on these graphs and the implied offer prices, is the Acquirer overpaying?

Maybe.

This Contribution Analysis excludes Synergies, which make a *big* difference; they would skew the numbers in favor of the Target if we credited them to the Target.

Also, this deal won't necessarily be 100% Stock; in the case study questions and presentation here, we recommended a mixed Cash, Debt, and Stock structure instead.

With a mixed Cash/Debt/Stock deal, the Target will own far less of the Combined Company, so this question about the correct ownership percentage is much less important.

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Value Creation Analysis

You can also evaluate a deal by assessing **its possible impact on the Acquirer's share price.**

In a previous section of this guide, we stated these two rules:

1. **Combined Equity Value** = Acquirer's Equity Value + Market Value of Stock Issued in Deal.

2. **Combined Enterprise Value** = Acquirer's Enterprise Value + Purchase Enterprise Value of Target.

These principles work for interview questions, **but they don't always hold up in real life**.

Immediately after an acquisition is announced, the Combined Equity Value and Combined Enterprise Value might change as described above – but after that, anything could happen.

For example, if the market doesn't "like" the deal, the Acquirer's share price could plummet.

Or if the deal turns out to be much *better* than expected, the Acquirer's share price could skyrocket, and the combined entity might end up being worth more than the Acquirer + Target.

To account for these possibilities, you can use a **Value Creation Analysis** to estimate the Combined Company's share price after the deal closes.

In a Value Creation Analysis, you start by finding larger companies in the same industry that trade at higher multiples because of greater scale, higher growth rates, a more favorable market position, or some other factor.

For this deal, we selected two larger companies in building material-adjacent sectors: Carrier Global Corporation [CARR] and W.W. Grainger, Inc. [GWW].

Each one had $10 – 20 billion of revenue, and each conveniently traded at higher Revenue and EBITDA multiples.

Then, we added up the Acquirer and Target's standalone figures, including Synergies in the first combined year, to put the financial profiles side-by-side.

For example, the FY 21 Projected EBITDA for the Acquirer + Target here equals each company's standalone EBITDA *plus* the net impact of Synergies and Merger & Integration Costs.

Depending on the assumptions, the Combined EBITDA might be *lower* in the first year or two since most of the integration costs are incurred then!

In the next step, we averaged the Revenue and EBITDA multiples of CARR and GWW and *applied these multiples* to the Combined Company:

Operating Statistics:								
	Equity	Enterprise		Revenue			EBITDA	
Company Name	Value	Value	LTM	FY20	FY21	LTM	FY20	FY21
Combined Entity - Post-Transaction and Acquisition Effects:			11,232.9	11,963.0	12,970.3	758.3	798.0	907.6

Valuation Statistics:								
	Equity	Enterprise		TEV / Revenue			TEV / EBITDA	
Company Name	Value	Value	LTM	FY20	FY21	LTM	FY20	FY21
Standalone Entities:								
Builders FirstSource, Inc.	3,323.6	4,559.1	0.6 x	0.6 x	0.5 x	9.0 x	8.6 x	7.5 x
BMC Stock Holdings	2,215.8	2,304.6	0.6 x	0.6 x	0.6 x	9.1 x	8.6 x	7.7 x
Closest Comparable Public Companies for Combined Entity:								
Carrier Global Corporation	26,878.2	36,541.2	2.1 x	2.2 x	2.1 x	14.3 x	15.6 x	13.3 x
W.W. Grainger, Inc.	19,375.9	21,332.9	1.8 x	1.8 x	1.8 x	15.0 x	14.1 x	12.9 x
Combined Entity - Post-Transaction and Acquisition Effects:			2.0 x	2.0 x	1.9 x	14.7 x	14.8 x	13.1 x
Combined Entity - Implied Enterprise Value:			22,244.3	24,291.9	24,980.0	11,108.6	11,838.0	11,917.2

Yes, we're assuming that the Combined Company will trade at significantly higher Revenue and EBITDA multiples because of… magic? Leprechauns? It's not entirely clear.

The Value Creation Analysis is *highly* speculative, which is why many people don't take it seriously.

It's useful in industries where larger companies *predictably* trade at higher multiples due to scale and geographic diversity (e.g., commercial banks and REITs), but it's a bit of a stretch to say it is highly relevant in the building materials sector.

Continuing with the analysis, we added and subtracted the normal bridge items to move from the Implied Enterprise Value to the Implied Equity Value.

We also had to include the Cash, Debt, and Stock used to fund the deal, which meant a lower combined Cash balance, a higher Debt balance, and a higher Share Count.

You can see an image summarizing this analysis on the next page.

The "Combined Entity – Implied Share Price" represents this Combined Implied Equity Value divided by (Acquirer's Share Count + Shares Issued to Fund the Deal).

The results for the Revenue multiples seem so unbelievable that we ignore them.

The results for the EBITDA multiples also show a substantial increase in the Acquirer's share price, but it's closer to the "believable zone."

Valuation Statistics:								
	Equity	Enterprise	TEV / Revenue			TEV / EBITDA		
Company Name	Value	Value	LTM	FY20	FY21	LTM	FY20	FY21
Standalone Entities:								
Builders FirstSource, Inc.	3,323.6	4,559.1	0.6 x	0.6 x	0.5 x	9.0 x	8.6 x	7.5 x
BMC Stock Holdings	2,215.8	2,304.6	0.6 x	0.6 x	0.6 x	9.1 x	8.6 x	7.7 x
Closest Comparable Public Companies for Combined Entity:								
Carrier Global Corporation	26,878.2	36,541.2	2.1 x	2.2 x	2.1 x	14.3 x	15.6 x	13.3 x
W.W. Grainger, Inc.	19,375.9	21,332.9	1.8 x	1.8 x	1.8 x	15.0 x	14.1 x	12.9 x
Combined Entity - Post-Transaction and Acquisition Effects:			2.0 x	2.0 x	1.9 x	14.7 x	14.8 x	13.1 x
Combined Entity - Implied Enterprise Value:			22,244.3	24,291.9	24,980.0	11,108.6	11,838.0	11,917.2
(+) Cash & Cash-Equivalents:			202.8	202.8	202.8	202.8	202.8	202.8
(+) Equity Investments & Non-Core Assets:			31.5	31.5	31.5	31.5	31.5	31.5
(-) Total Debt:			(3,189.9)	(3,189.9)	(3,189.9)	(3,189.9)	(3,189.9)	(3,189.9)
(-) Preferred Stock:			-	-	-	-	-	-
Combined Entity - Implied Equity Value:			**19,288.7**	**21,336.3**	**22,024.4**	**8,153.0**	**8,882.4**	**8,961.6**
Combined Entity - Diluted Share Count:			153.045	153.045	153.045	153.045	153.045	153.045
Combined Entity - Implied Share Price:			**$ 126.03**	**$ 139.41**	**$ 143.91**	**$ 53.27**	**$ 58.04**	**$ 58.56**
Buyer - Share Price Prior to Transaction:			27.91	27.91	27.91	27.91	27.91	27.91
Per-Share Value Created:			**98.12**	**111.50**	**116.00**	**25.36**	**30.13**	**30.65**
Per-Share Value of Equity Purchase Premium:			1.98	1.98	1.98	1.98	1.98	1.98

The Enterprise Value bridge must include the Cash and Debt used to fund this deal in addition to the Acquirer and Target's numbers. The Share Count should equal Acquirer's Shares + Shares Issued to Fund the Deal.

The conclusion is that if there's even modest multiple expansion – for example, if the Combined Company goes from a 7.5x forward TEV / EBITDA multiple to an 8.2x or 8.8x multiple – the Acquirer's share price will increase after the deal closes.

But if the Combined Multiple stays the same as the Acquirer or Target's standalone multiples, the Acquirer's share price will *decrease*.

Synergies also make a difference, but this entire analysis comes down to the likelihood of modest multiple expansion (we're skeptical, but who knows).

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Interview Questions

Merger models and M&A deals can get complex, but most interview questions about these topics are **simple** and relate to the rationale for deals and the basic calculations.

The interview questions in this section reflect that reality. You need to know the fundamentals **very well** because the "difficult questions" relate to the core topics rather than the more advanced points or supplemental analyses.

M&A Concepts and Overview

Questions about accretion/dilution and related calculations could come up, but you're also likely to get many **conceptual questions** about why deals happen.

1. Why might one company want to buy another company?

One company might want to buy another company if it believes it will be better off after the acquisition takes place. For example:

- The Seller's **asking price** is less than its **Implied Value**, i.e., the Present Value of its future cash flows.
- The expected **IRR** from the acquisition exceeds the Buyer's **Discount Rate**.

Buyers often acquire Sellers to save money via consolidation and economies of scale, to expand geographically or gain market share, to acquire new customers or distribution channels, and to expand their products and services.

Synergies, or the potential to reduce expenses through consolidation or boost revenue through additional sales, also motivate many deals.

2. How can you analyze an M&A deal and determine whether it makes sense?

The qualitative analysis depends on the factors above: Could the deal help the company expand geographies, products, or customer bases, give it more intellectual property, or improve its team?

The quantitative analysis might include a Seller valuation to see if it's undervalued and a comparison of the expected IRR to the Buyer's Discount Rate.

Finally, EPS accretion/dilution is important in most deals because Buyers prefer to execute accretive deals, i.e., ones that increase their Earnings per Share (EPS). The Board of Directors is more likely to approve of accretive deals, and investors also prefer accretive deals.

3. Walk me through a merger model (accretion/dilution analysis).

In a merger model, you start by projecting the financial statements of the Buyer and Seller. Then, you estimate the Purchase Price and the mix of Cash, Debt, and Stock used to fund the deal. You create a Sources & Uses schedule and Purchase Price Allocation schedule to estimate the after-effects of the deal on the financial statements.

Then, you combine the Balance Sheets of the Buyer and Seller, reflecting the Cash, Debt, and Stock used, new Goodwill created, and any write-ups and write-downs. You then combine the Income Statements, reflecting the Foregone Interest on Cash, Interest Paid on New Debt, and Synergies.

The Combined Net Income equals the Combined Pre-Tax Income times (1 – Buyer's Tax Rate), and the Combined EPS equals the Combined Net Income divided by (Buyer's Existing Share Count + New Shares Issued in the Deal).

The EPS accretion/dilution equals the percentage difference between this Combined EPS and the Buyer's standalone EPS.

4. Why might an M&A deal be accretive or dilutive?

A deal is accretive if the extra Pre-Tax Income from a Seller exceeds the cost of the acquisition in the form of the Foregone Interest on Cash, Interest Paid on New Debt, and New Shares Issued.

For example, if the Seller contributes $100 in Pre-Tax Income, but the deal costs the Buyer only $70 in additional Interest Expense, and the Buyer doesn't issue any new shares, the deal will be accretive because the Buyer's Earnings per Share (EPS) will increase.

A deal will be dilutive if the opposite happens. For example, if the Seller contributes $100 in Pre-Tax Income, but the deal costs the Buyer $130 in additional Interest Expense, and its share count remains the same, its EPS will decrease.

5. How can you tell whether an M&A deal will be accretive or dilutive?

You compare the Weighted Cost of Acquisition to the Seller's Yield at its Purchase Price.

- **Cost of Cash** = Foregone Interest Rate on Cash * (1 – Buyer's Tax Rate)
- **Cost of Debt** = Interest Rate on New Debt * (1 – Buyer's Tax Rate)
- **Cost of Stock** = Reciprocal of the Buyer's P / E multiple, i.e., Net Income / Equity Value.
- **Seller's Yield** = Reciprocal of the Seller's P / E multiple, calculated using the Purchase Equity Value.

Weighted Cost of Acquisition = % Cash Used * Cost of Cash + % Debt Used * Cost of Debt + % Stock Used * Cost of Stock.

If the Weighted Cost is **less than** the Seller's Yield, the deal will be **accretive**; if the Weighted Cost is **greater than** the Seller's Yield, the deal will be **dilutive**.

6. That sounds complicated. Are there any shortcuts for guesstimating whether an M&A deal will be accretive or dilutive?

If it is a **100% Stock deal**, you can compare the Buyer's P / E multiple to the Seller's P / E multiple *at* the purchase price. If the Buyer's multiple is higher, the deal will be accretive (since the reciprocals of the P / E multiples in a 100% Stock deal are the Weighted Cost of Acquisition and the Seller's Yield).

For example, if the Buyer's P / E multiple is 10x, the Weighted Cost of Acquisition is 10%.

If the Seller's Current Equity Value is $1000, the Buyer pays a 20% premium, and the Seller's Net Income is $200, its P / E multiple *at* the purchase price is $1200 / $200 = 6x.

Therefore, this deal is accretive because the Seller's Yield is 1 / 6 = 16.7%, which is higher than the Weighted Cost of Acquisition.

7. Why do you focus so much on EPS in M&A deals?

Because it's the only easy-to-calculate metric that also captures the **FULL** impact of the deal – the Foregone Interest on Cash, Interest Paid on New Debt, and New Shares Issued.

Metrics such as EBITDA and Unlevered FCF more accurately represent the value of the company's core business, but they don't reflect *the deal's full impact* because they exclude the Net Interest Expense and New Shares Issued.

8. How do you determine the Purchase Price in an M&A deal?

If the Seller is public, you assume a **premium** to the Seller's current share price based on the average premiums for similar deals in the market (usually between 10% and 30%). You can also use the DCF, Public Comps, and other valuation methodologies to cross-check this figure.

The Purchase Price for private Sellers is based on the standard valuation methodologies, and you usually base it on a multiple of EBITDA, EBIT, or Revenue since private companies don't have easy-to-determine share prices.

If the Buyer expects to realize significant Synergies, it is often willing to pay a higher premium for the Seller because the Present Value of the Synergies might exceed this premium.

9. What is the "true price" in an M&A deal: The Purchase Equity Value or Purchase Enterprise Value? Why?

The "true price" is the Purchase Enterprise Value (or something close to it) because that represents what the Buyer pays *all* the Seller's investors for its core assets. However, the Purchase Enterprise Value is *not* necessarily what the Buyer pays in "upfront capital."

Also, for analytical purposes, the Purchase Equity Value is often the key assumption and drives the Cash, Debt, and Stock used to fund a deal. The Purchase Equity Value is also what the selling shareholders receive in the deal.

10. Can you expand on that answer? Why are many M&A models based on the Purchase Equity Value if what you just said about the "true price" is correct?

The issue is that the Seller's existing Debt and Cash do not necessarily make a *significant net impact* on the Combined Income Statement, but the Cash, Debt, and Stock used to fund the Purchase Equity Value always do.

For example, if the Seller's existing Debt is refinanced and replaced in the deal, the interest rate may shift a little, but there won't be a dramatic difference in the Seller's Pre-Tax Income: It still deducts the Interest Expense on that Debt.

If the Seller has Excess Cash, its post-deal Interest Income will be lower if this Cash is distributed to its shareholders, but that will also reduce the effective purchase price for the Buyer, meaning that it uses less Cash/Debt/Stock to fund the deal. So, the Seller's existing Cash and Debt tend to "cancel out" or, at least, make a small difference.

11. OK, but what about cash-free, debt-free deals for private companies? Aren't they different since they're based on Purchase Enterprise Value?

No. They *look different* on paper, but these deals are economically equivalent if you go through the math.

For example, in a cash-free, debt-free deal, the Uses side of the Sources & Uses schedule is based on Purchase Enterprise Value (Purchase Equity Value – Cash + Debt), and there's a Minimum Cash line on the Uses side.

This might look different from the non-cash-free, debt-free version, but two components of it are the same (the Purchase Equity Value and Assume/Refinance Existing Debt are both on the Uses side).

The Seller's Cash netted against its Minimum Cash equals its Excess Cash; this is a negative on the Uses side in the CF/DF version but a positive on the Sources side in the non-CF/DF version. They're equivalent because a negative on the Uses side is a positive on the Sources side.

The Combined Income Statement also looks different, but it's also equivalent if you go through the math; it's just that line items are split up differently (e.g., Interest Expense might be the total rather than Interest on New Debt vs. Interest on Assumed/Refinanced Debt).

12. What are the advantages and disadvantages of each purchase method (Cash, Debt, and Stock) in M&A deals?

Cash tends to be the cheapest option; most companies earn little Interest Income on it, so they don't lose much by using it to fund deals. Cash-based deals are also fast and easy to close.

The downside is that using Cash limits the Buyer's flexibility if it needs the funds for something else soon.

Debt is normally cheaper than Stock but more expensive than Cash, and deals involving Debt take more time to close because of the need to market the new Debt issuance.

Debt also limits the Buyer's flexibility because additional Debt makes *future* Debt issuances more difficult and expensive and might limit the total amount that can be raised.

Stock tends to be the most expensive option, but it can sometimes be the cheapest (on paper) if the Buyer trades at an extremely high P / E multiple.

It dilutes the Buyer's existing investors but also prevents the Buyer from paying an additional *cash expense* for the deal.

In some cases, the Buyer can also issue Stock more quickly than it can issue Debt.

Finally, the tax implications also differ: The Seller's shareholders are not taxed immediately if they receive shares from the Buyer, but they do pay taxes immediately if they receive cash proceeds for their shares (in Cash and Debt deals).

13. How does an Acquirer determine the mix of Cash, Debt, and Stock to use in a deal?

Since Cash is the cheapest for most Acquirers, they'll use all they can before moving to the other funding sources. So, you might assume that the Cash Available equals the Acquirer's current Cash balance minus its Minimum Cash balance, also factoring in the Target's Cash and Minimum Cash when applicable.

After that, Debt tends to be the next cheapest option. An Acquirer might be able to raise Debt to the level where its Debt / EBITDA remains in-line with peer companies'.

So, if it's levered at 2x EBITDA now, and similar companies have 4-5x Debt / EBITDA, it might be able to raise an additional 2-3x EBITDA worth of Debt. Again, you may also factor in the Target's Debt and EBITDA if they are significant.

Finally, there's no strict limit on the amount of Stock an Acquirer might issue, but few companies would issue enough to give up control of the company, and some Acquirers will issue Stock only up to the point at which the deal turns dilutive.

14. Which purchase method does a Seller prefer in an M&A deal?

The Seller must balance **taxes** with the **payment certainty** and **potential future upside**.

To a Seller, Debt and Cash are similar because they mean immediate payment, but also immediate capital gains taxes for the shareholders and no potential upside if the Buyer's share price increases. But there's also no risk if the Buyer's share price decreases.

Stock is more of a gamble because the Seller could end up with a higher price if the Buyer's share price increases, but it could also get a lower price if the Buyer's share price drops. The Seller's shareholders also avoid immediate taxes with Stock since they pay taxes only when they sell their shares.

So, the preferred method depends on the Seller's confidence in the Buyer: Cash and Debt are better when the uncertainty is higher, while Stock may be better with large, stable Buyers.

15. What's the impact of each purchase method in an M&A deal, and how do you estimate the Cost of each method?

The Foregone Interest on Cash represents the Cost of Cash. The Acquirer loses future projected Interest Income by using Cash to fund a deal. The Interest Expense on New Debt represents the Cost of Debt.

For both, you multiply the interest rate by (1 – Acquirer's Tax Rate) to estimate the after-tax costs.

The Cost of Stock is represented by the additional shares created in a deal and how those shares reduce the Combined Company's EPS. It's the reciprocal of the Acquirer's P / E Multiple, i.e., the Acquirer's Net Income / Acquirer's Equity Value.

16. Isn't the Foregone Interest on Cash just an "opportunity cost"? Why do you include it?

No, it's **not** just an "opportunity cost" because the Acquirer's *projected* Pre-Tax Income already includes the Interest Income that the company *expects* to earn on its Cash balance!

So, if an Acquirer forecasts $90 in Operating Income and $10 in Interest Income for a total of $100 in Pre-Tax Income, its projected Pre-Tax Income will **fall** if it uses Cash to fund the deal.

17. Isn't it contradictory to calculate the Cost of Stock using the reciprocal of the Acquirer's P / E multiple? What about the Risk-Free Rate, Beta, and the Equity Risk Premium?

It's not a contradiction; it's just a different way of measuring the Cost of Equity.

The "Reciprocal of the P / E Multiple" method measures the Cost of Equity in terms of EPS impact, while the CAPM method (with the RFR, Beta, and ERP) measures it based on the stock's expected annualized returns.

Neither method is “correct” or “incorrect” because you use them in different contexts.

Regardless of the method, Stock is usually a company's most expensive funding source.

18. Why might an Acquirer choose to use Stock or Debt even if it could pay for the Target with 100% Cash?

The Acquirer might not necessarily use its entire Cash balance if much of it is in overseas subsidiaries or otherwise restricted.

Also, the Acquirer might be preserving its Cash for a future expansion plan, Debt maturity, or other purpose.

Finally, if the Acquirer trades at high multiples, such as a 100x P / E multiple, using Stock to fund the deal might be cheaper.

19. Are there cases where EPS accretion/dilution is NOT important? What other analyses could you look at?

Yes, there are many cases where EPS accretion/dilution is less important.

For example, if the Buyer is private or has negative EPS as a standalone entity, it won't care whether the deal is accretive or dilutive.

It also makes little difference if the Buyer is far bigger than the Seller (e.g., 10x – 100x its size).

Besides EPS accretion/dilution, you can also analyze the deal's qualitative merits, compare the IRR to the Discount Rate, and value the Seller plus the Synergies and compare that to the Equity Purchase Price.

Finally, you can create a Contribution Analysis to determine how much the Buyer and Seller "contribute" to each financial metric and then compare the contribution percentages to their respective ownership percentages, assuming it’s a 100% Stock deal.

Value Creation Analysis, to determine how the Buyer’s share price will change after the deal closes, may also be useful in certain contexts, such as if the Buyer + Seller together will resemble a larger, more valuable public company in the market.

20. How does a merger differ from an acquisition?

There are no mechanical differences because **there's always a Buyer and Seller in any M&A deal**.

The difference is that in a merger, the companies are closer in size, while the Buyer is significantly larger than the Seller in an acquisition.

100% Stock or majority-Stock deals are also more common in mergers because similarly sized companies can rarely use only Cash or Debt to acquire each other.

You'll also place more weight on the Contribution Analysis and Value Creation Analysis in mergers because of this prevalence of 100% Stock deals.

21. What are the main PROBLEMS or limitations of merger models?

First, EPS is not always a meaningful metric. Second, Net Income and "cash flow" are quite different, so EPS-accretive deals could be bad in terms of cash flows or valuation.

Third, merger models don't capture the true risk inherent in M&A deals; 100% Cash deals almost always look accretive, even though the integration process might go wrong, legal issues might arise, and customers or shareholders might revolt.

Fourth, merger models often fail to consider what might happen if the Buyer or Seller's share prices change significantly before the deal closes.

Finally, merger models don't capture the qualitative factors required for successful deals, such as the cultural fit or management's ability to work together.

22. You are an investment banker advising a client on a sell-side M&A deal. Can you walk me through the process and explain the main steps?

In a sell-side M&A deal, bankers market the company to potential buyers and help negotiate a deal for the best price and terms possible.

The bankers start by meeting the company, creating initial marketing materials like the Executive Summary and Offering Memorandum (OM), and deciding on potential buyers.

Then, they send the Executive Summary to potential buyers to gauge their interest, and they respond to follow-up requests for more information.

Next, they narrow the list of potential buyers by setting a bid deadline and requiring written Indications of Interest (IOIs) from buyers. Bankers select certain bidders to advance, continue to send them information, and narrow the list.

Eventually, they pick the winner, negotiate the Purchase Agreement, and announce the deal. Bankers try to maximize the client's price, but they also add value by negotiating the other terms and winning competing offers.

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Accretion/Dilution Calculations

Questions about accretion/dilution math can be surprisingly tricky and often turn into "gotchas," so it's best to practice with many examples.

1. Company A, with a P / E of 25x, acquires Company B for a purchase P / E multiple of 15x. Will the deal be accretive?

You can't tell unless you know it's a 100% Stock deal.

If it *is* a 100% Stock deal, then it will be accretive because the Buyer's P / E is higher than the Seller's, indicating that the Buyer's Cost of Acquisition (1 / 25, or 4%) is less than the Seller's Yield (1 / 15, or 6.7%).

2. Company A has 10 shares outstanding at a share price of $25.00, and its Net Income is $10.

It acquires Company B for a Purchase Equity Value of $150. Company B has a Net Income of $10 as well. Assume the same tax rates for both companies. How accretive is this deal?

Company A's EPS is $10 / 10 = $1.00.

Company A must issue 6 new shares to do the deal since $150 / $25.00 = 6, so the Combined Share Count is 10 + 6 = 16.

Since no Cash or Debt is used and the tax rates are the same, the Combined Net Income = Company A Net Income + Company B Net Income = $10 + $10 = $20.

Therefore, the Combined EPS is $20 / 16 = $1.25, so it's accretive by 25%.

3. Company A now uses Debt with an Interest Rate of 8% to acquire Company B. Is the deal still accretive? At what interest rate does it change from accretive to dilutive?

The Weighted Cost of Acquisition is 8% * (1 – 25%), or 6%, so the deal is still accretive because that Cost is less than the Seller's Yield of 6.7%.

For the deal to turn dilutive, the After-Tax Cost of Debt would have to exceed 6.7%. Since 6.7% / (1 – 25%) = 8.9%, the deal would turn dilutive at an interest rate >= 8.9%.

4. What are the Combined Equity Value and Enterprise Value in this deal?

Assume that Equity Value = Enterprise Value for both the Buyer and Seller and that the deal uses 100% Stock funding.

Combined Equity Value = Buyer's Equity Value + Market Value of Stock Issued in the Deal = $250 + $150 = $400.

Combined Enterprise Value = Buyer's Enterprise Value + Purchase Enterprise Value of Seller = $250 + $150 = $400.

5. How do the Combined TEV / EBITDA and P / E multiples change if the deal financing method changes?

The Combined TEV / EBITDA stays the same regardless of the financing method, but the Combined P / E multiple changes based on the Stock issued and the Cash and Debt used.

The Stock issued affects the Combined Equity Value, and the Cash and Debt used affect the Combined Net Income because of the Foregone Interest on Cash and Interest Paid on New Debt.

6. Without doing any math, what range would you expect for the Combined P / E multiple?

The Combined P / E multiple should be between the Buyer's P / E multiple and the Seller's Purchase P / E multiple, so it's between 25x and 15x here.

Since Company A is larger than Company B, we would expect the Combined P / E multiple to be closer to Company A's multiple of 25x.

7. Now assume that Company A is twice as big, so its Equity Value is $500, and its Net Income is $20. Will a 100% Stock deal be more or less accretive?

The deal will be *less accretive*.

The intuition is that Company A's P / E remains the same, but it's significantly bigger, so the higher-yielding Company B provides less of a boost to Company A's EPS.

The Combined P / E multiple will still be between 15x and 25x, but it will be even closer to 25x because Company A has a higher weighting in the combined company.

8. Now, do the math. What is the accretion/dilution in a 100% Stock deal with a $150 Purchase Equity Value for Company B? Feel free to write down the numbers.

Company A's share price is now $50.00, it still has 10 shares outstanding, and its Equity Value is $500. Its EPS is $20 / 10 = $2.00.

Company A must issue 3 additional shares to acquire Company B since $150 / $50.00 = 3.

Since both companies have the same tax rate and no Cash or Debt was used, you can add the Net Income figures: Combined Net Income = $20 + $10 = $30.

The new share count is 10 + 3 = 13, and the new EPS is $30 / 13 = $2.31. This is about 15% higher than the Buyer's standalone EPS.

So, it's ~10% lower than the 25% accretion when Company A was smaller.

9. Company A has a P / E of 10x, a Debt Interest Rate of 8%, a Cash Interest Rate of 4%, and a Tax Rate of 25%.

It wants to acquire Company B at a purchase P / E multiple of 16x using 1/3 Stock, 1/3 Debt, and 1/3 Cash. Will the deal be accretive?

Company A's After-Tax Cost of Stock is 1 / 10, or 10%, its After-Tax Cost of Debt is 8% * (1 – 25%) = 6%, and its After-Tax Cost of Cash is 4% * (1 – 25%) = 3%.

Company B's Yield is 1 / 16, or 6.25% (mental math: 1 / 4 = 25%, and 1 / 8 = 12.5%, so 1 / 16 = 6.25%).

The Weighted Cost of Acquisition is 10% * 1/3 + 6% * 1/3 + 3% * 1/3 = 3.33% + 2% + 1% = 6.33%.

The deal will be **dilutive** since the Weighted Cost is slightly above Company B's Yield.

10. Company A acquires Company B using 100% Debt. Company B has a purchase P / E multiple of 12x, and Company A has a P / E multiple of 15x.

What interest rate on Debt is required to make the deal dilutive?

Company B's Yield is 1 / 12, or 8.3%, so the After-Tax Cost of Debt must be *above* 8.3% for the deal to be dilutive.

Assuming the company has a tax rate of 25%, 8.3% / (1 – 25%) = 11.1%, which you can round to "Around 11%" (mental math: there's a 33% difference between 8% and 12%, but the tax rate is only 25% here, so the interest rate must be slightly less than 12%).

11% is a high interest rate for most companies, so a 100% Debt deal would probably be accretive.

11. Company A has an Equity Value of $1,000 and a Net Income of $100. Company B has a Purchase Equity Value of $2,000 and a Net Income of $50.

How much in Synergies must be realized for a 100% Stock deal to be accretive?

Company A's P / E is $1,000 / $100 = 10x, so its Cost of Stock is 10%. Company B's P / E is $2,000 / $50 = 40x, so its Yield is 1 / 40, or 2.5%.

Therefore, without Synergies, this deal would be highly dilutive.

Company B's Yield must exceed 10% for the deal to turn accretive. That means its Purchase P / E multiple must be below 10x, meaning its Net Income must be above $200.

So, there must be **$150 in After-Tax Synergies** for this deal to be accretive. At a 25% tax rate, that means at least $200 in Pre-Tax Synergies.

12. An Acquirer has an Equity Value of $1 billion, Cash of $50 million, EBITDA of $100 million, Net Income of $50 million, and a Debt / EBITDA of 2x. Peer companies have a median Debt / EBITDA of 4x.

It wants to acquire another company for a Purchase Equity Value of $500 million. The Seller has a Net Income of $30 million, EBITDA of $50 million, and no Debt.

What's the best way to fund this deal? Use the Combined EBITDA figures in the calculations.

The Acquirer would prefer to use its Cash balance to do this deal, but $50 million is probably close to its minimum Cash for a company of this size, so Cash financing is unlikely.

The company could afford to boost its Debt / EBITDA from 2x to 4x since peer companies have leverage in that range.

The Combined Company has $150 million in EBITDA, and 4 * $150 million = $600 million.

The Acquirer has $200 million in Debt, and the Target has no Debt, so the Acquirer could afford to issue $400 million in new Debt to fund the deal.

The remaining $100 million could be funded with Stock. If the Acquirer used part of its Cash or Target's Cash balance, this $100 million Stock portion would be reduced.

13. An Acquirer has an Equity Value of $500 million, Cash of $100 million, EBITDA of $50 million, Net Income of $25 million, and Debt / EBITDA of 3x.

Similar companies in the market have Debt / EBITDA ratios of 5x.

What's the BIGGEST acquisition this company might be able to complete?

You can't answer this question precisely without knowing the Target's Net Income and EBITDA, but you can make a rough estimate.

The Acquirer couldn't use its *entire* Cash balance to fund a deal, but it might be able to use a substantial portion of it, such as $50 million, since its remaining Cash balance would then equal its annual EBITDA.

It could afford to use leverage up to 5x EBITDA, which means it could use $100 million in additional Debt to fund a deal (since it currently has $150 million of Debt, or 3 * $50 million).

That number might also change based on the Target's Debt and EBITDA.

There's no limit to how much Stock the company could issue, but it probably wants to limit the issuance to a level that lets it retain control.

Therefore, the likely maximum is around $500 million of Stock, and a more realistic level might be about half its Current Equity Value ($250 million), or the amount that turns the deal dilutive.

So, a reasonable answer might be: "In theory, the Acquirer might be able to fund a deal for up to $50 + $100 + $500 = $650 million. But a more realistic maximum level would be around $400 million, assuming a $250 million Stock issuance."

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Equity Value, Enterprise Value, and Multiples in M&A Deals

Questions about Equity Value and Enterprise Value in M&A deals are less common than ones on accretion/dilution, but they often come up as follow-up questions to other scenarios.

1. An Acquirer with an Equity Value of $500 million and an Enterprise Value of $600 million buys another company for a Purchase Equity Value of $100 million and a Purchase Enterprise Value of $150 million.

What are the Combined Equity Value and Enterprise Value?

The Combined Enterprise Value equals the Enterprise Value of the Buyer plus the Purchase Enterprise Value of the Seller, so it's $600 + $150 = $750 million.

You can't determine the Combined Equity Value because it depends on the deal financing: Combined Equity Value = Acquirer's Equity Value + Market Value of Stock Issued in Deal.

If it's a 100% Stock deal, the Combined Equity Value will be $500 + $100 = $600 million, but if it's 100% Cash or Debt, the Combined Equity Value = $500 million.

Depending on the % Stock used, the Combined Equity Value will be between $500 and $600 million.

2. How do the Combined Equity Value and Enterprise Value change based on the deal financing?

The Combined Enterprise Value is not affected by the deal financing: It always equals the Buyer's Enterprise Value plus the Purchase Enterprise Value of the Seller.

The Combined Equity Value equals the Buyer's Equity Value plus the Market Value of Stock Issued in the Deal, which could range from $0 up to the Seller's Purchase Equity Value.

So, in a 100% Stock deal, the Combined Equity Value = Buyer's Equity Value + Purchase Equity Value of Seller.

3. Wait, you're saying that in a 100% Cash or Debt deal, the Seller's Equity Value just "disappears." How is that possible?

The Seller's Equity Value doesn't "disappear" – it's *transformed* into the Cash used or Debt issued by the Buyer in the deal.

The Combined Enterprise Value calculation demonstrates this point: *Both companies'* Enterprise Values still exist after the deal, so no value is "lost" along the way.

4. Wait a minute, you're also saying that the purchase premium the Acquirer pays for the Target will last after the deal closes? How is that possible?

The purchase premium does not necessarily "last" because it depends on the market's reaction to the deal.

If the market believes the premium was justified, the rules about the Combined Equity Value and Combined Enterprise Value will hold up.

However, if the market believes the Acquirer overpaid for the Target, the Acquirer's share price will fall to reflect the amount by which it overpaid.

5. Let's say an Acquirer has an Equity Value of $500 million and an Enterprise Value of $600 million. The Acquirer has 100 million shares outstanding at $5.00 per share.

The Target has an Equity Value of $100 million and an Enterprise Value of $150 million, and the Acquirer pays a 30% premium to acquire the Target in a 100% Stock deal.

A few months after the deal is announced, the market loses faith in the deal and believes the 30% premium is no longer justified.

What happens to the Combined Equity Value and Enterprise Value immediately after the deal is announced vs. several months later when the market loses faith in the 30% premium?

Immediately after, Combined Equity Value = $500 million + $130 million = $630 million since it's a 100% Stock deal.

Combined Enterprise Value = $600 million + $180 million = $780 million.

When the market loses faith in this 30% premium, the Acquirer's share price will fall such that its Equity Value and Enterprise Value both decrease by $30 million.

$30 million / 100 million shares = $0.30, so the Acquirer's share price will fall to $4.70, and the Combined Equity Value will decrease to $600 million because the Acquirer's Equity Value is now only $470 million.

Combined Enterprise Value = $570 million + $180 million = $750 million, so it is also down by this $30 million premium.

6. How does that last answer change if the Acquirer uses 100% Debt or Cash instead?

The Combined Enterprise Value changes the same way in both steps: Initially, it's $780 million, but it falls to $750 million when the Acquirer's share price falls.

The Combined Equity Value is initially only $500 million in a 100% Debt or 100% Cash deal because no Stock is issued. When the Acquirer's share price falls, the "Combined Equity Value," which is just the Acquirer's Equity Value here, drops to $470 million.

7. An Acquirer with an Equity Value of $500 million and an Enterprise Value of $600 million has a Net Income of $50 million and an EBITDA of $100 million.

The Target, with a Purchase Equity Value of $100 million and a Purchase Enterprise Value of $150 million, has a Net Income of $10 million and an EBITDA of $15 million.

What are the Combined P / E and TEV / EBITDA multiples in a 100% Stock deal? Assume the same tax rates for the Acquirer and Target.

The Combined Equity Value in a 100% Stock deal is $500 + $100 = $600 million, and the Combined Enterprise Value is $600 + $150 = $750 million.

The Combined EBITDA is $115 million, and the Combined Net Income, assuming the same tax rates and no interest effects since it's a 100% Stock deal, is $50 + $10 = $60 million.

Therefore, the Combined P / E multiple is $600 million / $60 million = 10x, and the Combined TEV / EBITDA multiple is $750 million / $115 million = ~6.5x.

8. How would those Combined Multiples change in a 100% Cash or Debt deal?

The Combined TEV / EBITDA multiple would stay the same because neither the Combined Enterprise Value nor the Combined EBITDA is affected by the deal financing.

The Combined P / E multiple would change because the Combined Equity Value would be only $500 million in a 100% Cash or Debt deal.

The Combined Net Income would also change because of the Foregone Interest on Cash and Interest Paid on New Debt.

In *most cases*, the Combined P / E multiple will be lower in a 100% Cash or 100% Debt deal because the Combined Equity Value will decrease by a greater percentage than the Combined Net Income.

9. How do the Combined Multiples change based on the deal financing?

Enterprise Value-based multiples do not change based on the % Cash, Debt, and Stock used because the Combined Enterprise Value is not affected by the deal financing, and TEV-based metrics such as Revenue, EBITDA, and EBIT are also not affected by it.

Equity Value-based multiples *do* change based on the deal financing because the Combined Equity Value depends on the Stock Used, and Equity Value-based metrics such as Net Income are affected by the Foregone Interest on Cash and Interest Paid on New Debt.

10. What are the possible ranges for the Combined Multiples after a deal takes place?

Combined Enterprise Value-based Multiples will be between the Buyer's standalone multiples and the Seller's purchase multiples.

Combined Equity Value-based Multiples are often in that range as well, but they do not have to be (see the next question for an example).

You **cannot** average the Buyer's multiples and the Seller's purchase multiples to determine the Combined Multiples because the companies could be different sizes.

You also **cannot** use a weighted average because the *proportions* of Enterprise Value, EBITDA, and other financial metrics from each company might be different.

The Combined Multiples will be closer to the Buyer's multiples if the Buyer is much bigger, but they'll be in the middle of the range if the Buyer and Seller are closer in size.

11. Consider this M&A scenario:

- **Company A: Enterprise Value of $100, Equity Value of $80, EBITDA of $10, Net Income of $4, and Tax Rate of 50%.**
- **Company B: Enterprise Value of $40, Equity Value of $40, EBITDA of $8, Net Income of $2, and Tax Rate of 50%.**

Calculate the TEV / EBITDA and P / E multiples for each company.

Company A TEV / EBITDA = $100 / $10 = 10x; P / E = $80 / $4 = 20x.

Company B TEV / EBITDA = $40 / $8 = 5x; P / E = $40 / $2 = 20x.

12. Company A acquires Company B using 100% Cash and pays no premium. Assume a 5% Foregone Interest Rate on Cash.

What are the Combined TEV / EBITDA and P / E multiples?

Combined TEV / EBITDA = Combined Enterprise Value / Combined EBITDA = $140 / $18 = ~7.8x.

Combined P / E = Combined Equity Value / Combined Net Income.

The Combined Equity Value is the Acquirer's Equity Value of $80 since no Stock was issued.

We can add both companies' Net Incomes since they have the same tax rate, so the Combined Net Income is $6. But we have to adjust for the Foregone Interest on Cash as well.

The Acquirer used $40 in Cash, and 5% * $40 = $2. After the 50% tax rate, that's a $1 reduction in Net Income.

So, the Combined Net Income is $5, which makes the Combined P / E = $80 / $5 = 16x.

13. Now, let's say that Company A uses 100% Debt with a 10% interest rate to acquire Company B.

Again, Company A pays no premium for Company B. What are the combined multiples?

The Combined TEV / EBITDA multiple remains the same at ~7.8x because the deal financing does not affect it.

The Combined Equity Value is still the Acquirer's Equity Value of $80.

The Combined Net Income before adjustments is $6, but now we must adjust for the Interest Paid on New Debt.

If Company A uses $40 of Debt to acquire Company B, it will pay $40 * 10% * (1 – 50%), or $2, in After-Tax Interest.

So, the Combined Net Income is $4, which makes the Combined P / E = $80 / $4 = 20x.

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Full Merger Model Mechanics

Ironically, questions about the **full merger model** tend to be easier than those in the previous sections because building a merger model is straightforward and doesn't lend itself to "mental math."

So, as with the first section, these questions tend to be more **conceptual**.

1. What information do you need from the Buyer and Seller to create a full merger model?

At the minimum, you need Income Statement projections for both companies over the next few years. Ideally, you will also create simple cash flow projections that track the changes in each company's Cash and Debt over the same period.

You do **not** need full 3-statement projections for both companies – like a DCF analysis, cash flow estimates without Balance Sheet projections are fine.

2. Why is a Sources & Uses schedule important in a full merger model?

The Sources & Uses (S&U) schedule spells out the deal financing in detail, including the exact amounts of Cash, Stock, and Debt used to fund the deal and what they're paying for (e.g., the Seller's shares, fees, refinanced Debt, etc.).

Without a S&U schedule, it's more difficult to calculate lines like the Foregone Interest on Cash in a full model because the total "Cash Used" might be spread across different assumptions.

3. How do the assumptions for a cash-free, debt-free deal for a private Seller differ from those of a standard M&A deal for a public Seller?

In a cash-free, debt-free deal, the Seller's existing Cash and Debt balances go to $0 when the deal closes and are **replaced with new Cash and Debt balances**.

The Cash is brought up to the Seller's Minimum Cash, and the new Debt balance is usually the same as the old one because it is simply replaced with a new issuance.

If Debt > Cash, the Seller uses its Cash balance to repay as much Debt as it can, and the remaining amount of Debt is deducted from the proceeds that go to the selling shareholders (i.e., they earn less because they must use some of their money to repay the Debt).

If Cash > Debt, the Seller repays its entire Debt balance using its Cash, and it distributes the remaining Cash to shareholders right before/as the deal closes, which reduces its Equity Value.

In these types of deals, the purchase price is usually based on a multiple such as TEV / EBITDA or TEV / Revenue rather than a share-price premium because the Seller is private.

Also, the Uses side of the Sources & Uses schedule is based on the Purchase Enterprise Value, the Seller's Minimum Cash, and the Transaction/Financing Fees.

The Sources side is standard and includes the usual Cash, Debt, and Stock line items.

4. What's the purpose of a Purchase Price Allocation schedule in a merger model?

The main purpose is to estimate the **Goodwill** created in a deal.

Goodwill exists because the Purchase Equity Value in deals almost always exceeds the Seller's Common Shareholders' Equity (CSE).

When this happens, the Combined Balance Sheet will go out of balance because the Seller's CSE is written down to $0, but the total amount of Cash, Debt, and Stock used in the deal is greater than the CSE that was written down. Goodwill exists to "plug the gap" and ensure the Balance Sheet balances.

So, you estimate the new Goodwill with this schedule, factor in write-ups of Assets such as PP&E and Intangibles, and include other acquisition effects such as the creation of a new Deferred Tax Liability and changes to existing Deferred Tax items.

5. Why do Deferred Tax Liabilities get created in many M&A deals?

A Deferred Tax Liability, or DTL, represents the *expectation* that Cash Taxes will exceed Book Taxes in the future.

DTLs get created because the Depreciation & Amortization on Asset Write-Ups is not deductible for cash-tax purposes in a Stock Purchase (i.e., an M&A deal where the Buyer purchases all the Seller's common shares and acquires *everything* the Seller has).

As a result, the Buyer will pay more in Cash Taxes than Book Taxes until the Write-Ups are fully depreciated/amortized. Each time the Buyer pays more in Cash Taxes than Book Taxes, the DTL decreases until it eventually reaches 0.

6. An Acquirer purchases a Target for a $1 billion Equity Purchase Price. This Target has $600 million in Common Shareholders' Equity and no existing Goodwill.

The Acquirer plans to write up the Target's PP&E and Other Intangible Assets by $100 million.

Walk me through the Purchase Price Allocation, assuming a 25% tax rate.

The "Allocable Purchase Premium" equals the Equity Purchase Price minus the Common Shareholders' Equity plus the Target's existing Goodwill, so $1 billion – $600 million + $0 = $400 million.

The PP&E and Other Intangible Assets increase by $100 million, so you *subtract* this figure because it means you'll need less Goodwill to make the Balance Sheet balance. So, the Purchase Premium is down to $300 million.

Then, you create a Deferred Tax Liability due to the future D&A on these asset write-ups. It's equal to $100 million * 25%, or $25 million, and you *add it* because an additional item on the L&E side means that more Goodwill will be needed on the Assets side.

So, $325 million of Goodwill gets created, along with Asset Write-Ups of $100 million and a new Deferred Tax Liability of $25 million.

7. What happens if an Acquirer purchases another company for a $1 billion Equity Purchase Price, but the Target's Common Shareholders' Equity is $1.5 billion?

Assume there are no write-ups or other adjustments.

"Negative Goodwill" cannot exist per the rules of IFRS and U.S. GAAP.

So, you record this $500 million difference as an Extraordinary Gain on the Income Statement, which increases Pre-Tax Income and Net Income.

On the Cash Flow Statement, Net Income is higher, and you reverse this Extraordinary Gain because it's non-cash. You also reverse the additional Book Taxes paid on it via a positive adjustment in the Deferred Taxes line item on the CFS.

The initial Balance Sheet combination still works the same way, but you don't record any Goodwill; you just add all the Target's Assets and Liabilities to the Acquirer's and reflect the Cash, Stock, and Debt used to fund the deal.

The increased Net Income (due to the Extraordinary Gain) flows into Common Shareholders' Equity, and the DTL changes based on the Deferred Tax line item adjustment.

Cash does not change because the Extraordinary Gain is non-cash, and the company's Cash Taxes stay the same.

8. What are the main adjustments when combining the Balance Sheets in an M&A deal?

You reflect the Cash, Debt, and Stock used in the deal, create new Goodwill, write up Assets such as PP&E and Other Intangibles, and reflect the Seller's assumed or refinanced Debt. You also show any new Deferred Tax Liabilities and the write-offs of existing DTLs and DTAs.

Then, you write down the Seller's Common Shareholders' Equity, deduct transaction fees from CSE, and deduct financing fees from the Book Value of the New Debt.

These are the most common adjustments, but there are others; for example, you might reduce the combined Accounts Receivable or Accounts Payable to reflect intercompany receivables or payables.

9. Give me an example of how you might estimate the Revenue and Expense Synergies in an M&A deal.

With Revenue Synergies, you might assume that the Seller can sell its products to some of the Buyer's customer base.

So, if the Buyer has 100,000 customers, 1,000 of them might buy widgets from the Seller. Each widget costs $10.00, which is $10,000 in extra Revenue.

There will also be COGS and Operating Expenses associated with these extra sales, so you must also factor those in. For example, if each widget costs $5.00, the Combined Company will earn only $5,000 in extra Pre-Tax Income.

With Expense Synergies, you might assume that the Combined Company can close a certain number of offices or lay off redundant employees, particularly in IT, accounting, and HR functions.

For example, if the Combined Company has 10 offices, management might believe only 8 will be required after the merger.

If each office costs $100,000 per year, there will be 2 * $100,000 = $200,000 in Expense Synergies, boosting the Combined Pre-Tax Income by $200,000.

10. Why do many merger models tend to overstate the impact of Synergies?

First, many merger models do **not** include the costs associated with Revenue Synergies. Even if the Buyer or Seller can sell more products or services after the deal occurs, those extra sales **cost something**, so you must include the extra COGS and OpEx.

Second, realizing Synergies **takes time**. Even if a company expects $10 million in "long-term synergies," it won't realize all of them in Year 1; it might take several years, and the percentage realized will increase gradually over time.

Finally, realizing Synergies **costs money**. There will always be "integration costs" associated with a deal, and certain types of Synergies, such as headcount reductions, will cost even more due to severance costs for employees.

11. How do you calculate the Combined Company's Debt repayment capacity in a merger model?

You do this by creating a "mini-Cash Flow Statement."

You eliminate most of the Financing and Investing sections (except for CapEx and potentially Dividends) but keep most of the Cash Flow from Operations section.

It's like what you do in a DCF to project Unlevered Free Cash Flow, but you're projecting the company's *Free Cash Flow* – which deducts Net Interest Expense – here.

You must include the Net Interest Expense because it directly affects a company's ability to repay Debt principal and generate Cash; it serves a different purpose than a DCF since you're tracking a company's Cash and Debt balances rather than valuing it.

12. How should you treat Stock-Based Compensation (SBC) in a merger model?

The easiest approach is to count it as a cash operating expense. Just as in a DCF, SBC is problematic because it increases the company's diluted share count and, therefore, reduces its value to existing shareholders.

But it's difficult to estimate this impact since you would have to project the company's share price and details of the SBC to do that. Also, it's much easier to analyze M&A deals if each company's standalone share count stays the same each year.

So, it's easiest to keep SBC as a deduction on the Income Statement (whether embedded within other expenses or separate) and **not** add it back on the Cash Flow Statement.

That way, it's effectively a cash operating expense, and each company's share count stays the same (assuming the Stock Issuances and Repurchases are also set to 0).

13. Why might you calculate metrics such as Debt / EBITDA and EBITDA / Interest for the Combined Company in an M&A deal?

These metrics tell you whether the Acquirer could use *more* Debt to fund the deal or if it's using *too much* Debt to fund the deal.

Sometimes, looking at a number like Debt / EBITDA immediately after a deal closes is deceptive because the Combined Company can de-lever rapidly by paying off Debt.

So, even if its Debt / EBITDA jumps up to a high level, such as 5x or 6x, if it can repay Debt and bring it down to 2x or 3x in a few years, it might be able to use more Debt to fund the initial deal.

14. How do Pro-Forma EPS and Pro-Forma accretion/dilution from the standard or IFRS/GAAP-compliant figures?

This point is confusing because there's no universal definition of Pro-Forma EPS.

But most companies calculate it by adding back non-cash expenses created in an M&A deal, primarily the Amortization of Intangibles and the Depreciation of PP&E Write-Ups, and some also add back Restructuring and Merger/Integration Costs, claiming they are "non-recurring."

Then, they calculate the Combined Net Income based on this "Pro-Forma" Pre-Tax Income with these line items added back.

Many companies report Pro-Forma EPS accretion/dilution based on these figures, but you should be skeptical because these numbers tend to understate the true costs of acquisitions.

15. Suppose that you set up an IRR vs. Discount Rate analysis to judge the merits of an M&A deal. What are its possible drawbacks and limitations?

One problem with this analysis is that it's unclear whether you should use the Buyer's Discount Rate, the Seller's, or a weighted average. Therefore, if the Buyer and Seller's Discount Rates differ significantly, the results could change dramatically.

Other problems relate to the treatment of Synergies and the Terminal Value of the Synergies. The IRR could change significantly depending on whether the Buyer or Seller gets "credit" for the Synergies.

There are also questions about whether a "Terminal Value" for Synergies is justified, as they probably won't last forever. Like EPS accretion/dilution, the IRR vs. Discount Rate analysis is useful, but it's still only one way to judge M&A deals.

16. Walk me through a Contribution Analysis for a 100% Stock M&A deal.

In a Contribution Analysis, you add the Buyer and Seller's financial metrics, such as Revenue and EBITDA, and determine the percentage the Buyer and Seller "contribute" to each combined metric.

Then, you estimate the Pro-Forma Combined Enterprise Value based on the Buyer's Enterprise Value / Buyer's Contribution Percentage for the relevant metric.

For example, if the Buyer's Enterprise Value is $1,500, and it contributes 75% of the Combined Revenue, the Pro-Forma Combined Enterprise Value based on Revenue is $1,500 / 75% = $2,000.

You then subtract the Buyer's Enterprise Value from this number to get the Seller's Implied Enterprise Value, and you subtract the items in the TEV bridge to get its Implied Equity Value.

Then, you divide by its share count to get the Implied Offer Price. You can then compare this Implied Offer Price to the actual Offer Price in the deal to determine whether the Buyer is paying an appropriate price.

17. How does the Value Creation Analysis in M&A deals work, and when is it appropriate?

In a Value Creation Analysis, you assume that the Buyer + Seller as a combined entity will trade at higher valuation multiples, in-line with the multiples of larger public companies in the sector.

You calculate the Combined Enterprise Value based on those higher multiples, subtract all the TEV bridge items, and reflect the Cash and Debt used in the deal to get the Combined Equity Value.

The Implied Share Price equals the Combined Equity Value / (Buyer's Share Count + Shares Issued in Deal).

If this share price exceeds the Buyer's standalone share price, the deal "created value."

This analysis is **highly speculative** because there's no guarantee that the Buyer + Seller combined will magically trade at higher multiples; it's most relevant if the deal represents a clear case of Companies #2 and #3 in the market combining to compete with Company #1.

It's less relevant when the market is highly fragmented or when the Buyer + Seller together still do not resemble larger companies.

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